



Open Chiplet Atlas Harness (OCAH)

Integrator Guide

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This guide provides instructions and best practices for integrating OCAH subsystems into your chiplet design. Each section covers integration considerations, interfaces, and configuration requirements for the respective subsystem.

Revision History

This document tracks major revisions and updates to the OCAH Integrator Guide.

Table 1. Revision History

Version	Date	Author	Description
0.0.1	2026-06-17	Alessandro Ottaviano	Initial integrator guide draft

Chapter 1. SMU Top-Level Integration

The System Management Unit (SMU) comprises three peer subsystems: the System Management Controller (SMC), the Debug & Test Ports (DTP), and the Security Processor (SEP). The subsections below describe each subsystem's interfaces independently, followed by the [SMU Composition](#) section that explains how they are wired together inside the SMU and how to select between the `smu.sv` logic-level core and the `smu_wrapper.sv` reference integration.

1.1. System Management Controller (SMC) Integration

The System Management Controller (SMC) is a subsystem responsible for power management, clock control, reset sequencing, and peripheral interface coordination. The `smc.sv` module exposes logic-level interfaces; all technology-specific components (PLLs, PVT sensors, eFuses, memory macros, GPIO padings) are instantiated externally by the adopter. See the [SMU Module Variants](#) section for how the reference wrappers demonstrate this integration.

For detailed architectural documentation about the SMC internal design, refer to `hw/smc/doc/`. The testbench `dv/smc/tb/tb_uvm/sv/smc_chiplet_wrap_uvm_top.sv` provides a complete integration example.

1.1.1. Clock and Reset

The SMC requires three external clock inputs: `clk_smc_i` (primary system clock driving the CPU cluster, fabric, and peripherals), `clk_periph_i` (peripheral clock driving the peripherals), and `clk_ref_i` (reference clock for timing-sensitive operations). All must be generated externally by the adopter's PLL infrastructure; the SMC provides AXI-Lite control outputs for PLL configuration (see [Peripheral Interface Connections](#)). The peripheral clock is used to drive the I3C interface, I2C interface, UART interface, and AVSBus interface.

Assert `powergood_i` after power-up sequence completion. Deassert `rst_cold_ni` after the power-up sequence completion.

The SMC generates AASD (async-assert / sync-deassert) reset outputs: `rst_primary_smc_clk_no` for the main clock domain, `rst_primary_periph_clk_no` for the peripheral clock domain, and `rst_primary_ref_clk_no` for the reference clock domain. These ensure clean reset release without metastability across the system hierarchy.

Optional reset override inputs (`rst_cold_override_ni`, `rst_cool_override_ni`, `rst_core_override_ni`) allow external control of specific reset domains for test and debug scenarios. Most integrations leave these override enables deasserted; the DTP drives them through JTAG IC_RESET controls when enabled (see [SMC-DTP Internal Interconnection](#)).

1.1.2. Address Map and Aperture

The SMC has a chiplet-visible aperture and an internal local address view. Internally, the SMC uses a fixed local base address of `0xC000_0000` and a fixed local window size of 16 MiB. SMC-local addresses (registers, memories, remapped regions) are defined as offsets from that base. The chiplet-visible aperture is adopter-defined: the chiplet or SoC address map assigns the global base and size that external masters can use to reach SMC through the SMU crossbar.

When integrating the SMC into a chiplet or SoC, the adopter assigns a system base address and reserved size for the SMC window. The system base (e.g. `0x0800_0000` in a chiplet address map) may be different from the SMC-local base `0xC000_0000`. Firmware programs the SMC `GLOBAL_BASE` / `REGION_SIZE` CSRs during early boot to match that assignment. The adopter's fabric or address decoder must route the reserved system range to the SMC block so that offsets are preserved: system base + offset maps to SMC-local `0xC000_0000` + offset.

Table 2. Address Base Mapping

View	Base Address	Description
SMC-local (internal)	0xC000_0000	Local alias address (fixed). Used by SMC CPU, firmware, and fabric.
System (adopter)	Adopter-defined	Base of the chiplet/SoC aperture assigned to SMC.

The SMC fabric uses three registers in the CPU Control block (0xC001_0000) to distinguish local from remote addresses:

- **LOCAL_BASE** (read-only, fixed at 0xC000_0000): The local alias base. Together with the fixed 16 MiB **LOCAL_ALIAS_REGION_SIZE**, this is the address space the SMC CPU and internal fabric use to access SMC resources.
- **GLOBAL_BASE** (read-write, defaults to 0x4000_0000): The system-level base address. Firmware must program this during early boot to match the adopter's system address assignment, enabling the SMC to issue outbound transactions using global system addresses.
- **REGION_SIZE** (read-write, currently resets to 0x0100_0000 / 16 MiB): The size of the SMC global aperture used by SMU-level routing and SEP-to-SMC decode. The SMC input fabric's local/global split uses the fixed **LOCAL_ALIAS_REGION_SIZE** parameter in RTL.

For full details on alias remapping and outbound routing, see the SMC Fabric documentation ([hw/smc/doc/fabric.adoc](#)).

For a detailed breakdown of all components within the SMC local map, refer to the SMC Memory Map documentation ([hw/smc/doc/memmap.adoc](#)).

1.1.3. AXI Interface Configuration

The SMC provides three distinct AXI interface groups serving different access paths. The **sys_axi_in** interface accepts requests from external system masters requiring access to SMC-managed resources. This interface uses a 56-bit address space with 64-bit data width and 6-bit transaction IDs. Connect this interface to your system fabric for normal operational access from application processors or DMA controllers.

The **sep_axi_in** interface provides dedicated access from the Security Processor (SEP) subsystem. This interface shares the same sizing parameters as the system input but operates independently to maintain security domain separation. Use this interface only when integrating alongside the SEP subsystem. The interface may remain unconnected if your integration does not include SEP.

The **output_axi** interface carries filtered and remapped requests from the SMC to external system resources. This interface uses an 8-bit transaction ID field to accommodate merged traffic from multiple input paths. Connect this interface to your system interconnect to allow SMC components access to external memory and peripherals. The testbench shows example connections using AXI VIP components that can be replaced with your system fabric.

The **jtag_axi_in** interface provides debug access through JTAG mechanisms. This interface uses reduced 2-bit transaction IDs and connects to your JTAG debug infrastructure. The testbench demonstrates typical debug access patterns through this interface.

1.1.4. GPIO Pad Integration

GPIO pad integration requires careful attention to pull-up/pull-down requirements and strap configurations. The GPIO table below specifies default directions and pull settings for each pin. During power-up, certain GPIO pins function as configuration straps that determine boot behavior and operational modes.

The testbench provides reference implementations for pad connections starting at line 732. For input-only pins such as the JTAG interface (pins 68-77), connect directly to external pads without output drivers. For bidirectional pins, implement proper tri-state buffers controlled by the **gpio_out_en_o** signals. The testbench demonstrates pull-up resistor

integration using `pullup` primitives on specific signals like I2C and I3C interfaces.

Configuration straps such as `BYPASS_SRAM_REPAIR` (pin 13), `TEST_EN` (pin 14), and others listed in the GPIO table must be set before the cold reset deassertion. These strap values determine fundamental operational modes and cannot be changed after the SMC completes fuse sensing. The testbench shows examples of strap forcing through GPIO pad connections.

For peripheral interfaces like UART, I2C, I3C, and SPI that multiplex through GPIO pins, the testbench demonstrates loopback configurations between transmit and receive pairs. In actual integration, connect these pins to external devices according to the pin function assignments shown in the GPIO table. The `lsio_interface_select_o` signals control peripheral multiplexing and are driven by SMC configuration registers.

1.1.5. Peripheral Interface Connections

The SMC exposes logic-level interfaces to several technology-specific subsystem components that the adopter instantiates externally. The subsections below describe each connection in turn.

PLL Interface

The `axil_pll_req_o` and `axil_pll_resp_i` signals form a 32-bit AXI-Lite interface to the adopter's PLL control logic. The SMC uses this interface to configure output frequencies and manage clock generation; connect it to the control/status registers of the external PLL block(s).

PVT Interface

The `axil_pvt_req_o` and `axil_pvt_resp_i` signals form a 32-bit AXI-Lite interface to the adopter's Process-Voltage-Temperature (PVT) monitoring blocks. Connect it to the external PVT sensor control registers to enable runtime monitoring and reporting.

eFuse Interface

The eFuse interface consists of two parts:

- `efuse_bank_ctrl` — a 32-bit AXI-Lite interface used to access the SHIM control CSRs (for example, foundry-specific bank timing such as the initialization delay before sensing can begin).
- `efuse_shim_command` — the custom fuse command interface used to issue read, program, and program/read-back operations to the eFuse bank.

Adopter responsibility. The `efuse_shim_command` interface is intentionally generic and technology-neutral. The adopter must implement an Interface SHIM that converts these generic commands into the hardware signals required by their OTP macro. The conversion is driven by the adopter's own read and program state machines inside the SHIM; OCAH only defines the command/response contract, not the macro-side signaling.

The command and response payloads are exchanged as packed structs:

```
// Controller -> SHIM
typedef struct packed {
    addr_t          address;           // Target address (in bits)
    logic            program_data;      // Data to program (when programming)
    access_length_words_t access_length_words; // Number of WORD_WIDTH words to access
    command_t        command;          // READ / PROGRAM / PROGRAM_READ_BACK
```

```

    logic                valid;                // Request valid
} fuse_command_req_t;

// SHIM -> Controller
typedef struct packed {
    data_t data;    // Read data (one WORD_WIDTH word)
    logic  status;  // Operation status (0 = OK, 1 = error)
    logic  valid;   // Response valid (qualifies `data` for one cycle)
} fuse_command_resp_t;

```

Fuse sensing protocol. During automatic sensing the controller issues a single read with `access_length_words` set to the entire bank (all `NUM_WORDS` words). The SHIM is expected to stream the bank back one `WORD_WIDTH` word at a time: for each OTP access it drives the read `data` onto `fuse_command_resp_o.data` and asserts `valid` for **exactly one cycle**. The controller consumes one word per cycle that `valid` is high, advancing to the next eFuse word on each assertion.

WARNING

`valid` must be asserted for only a single cycle per returned word. If the SHIM holds `valid` high for two cycles, the controller will sample the same `data` value a second time and count it as the data for the **next** eFuse word, shifting all subsequent words and corrupting the sensed shadow contents. The adopter's SHIM FSM is responsible for producing a clean one-cycle `valid` strobe per `WORD_WIDTH` access.

The `shadow_regs_o` output provides read-only access to the sensed fuse values. During bring-up the SMC automatically senses the fuse contents and propagates them through the shadow registers; the `fuse_sense_done_o` signal indicates completion of this sensing operation.

GPIO Interface

For GPIO controller access, connect the `axil_req_gpio_ctrl_o` and `axil_resp_gpio_ctrl_i` signals to the GPIO padding configuration logic. This interface allows runtime reconfiguration of pin directions, pull settings, and peripheral multiplexing options.

1.1.6. Memory Subsystem Integration

The SMC includes an embedded RISC-V CPU cluster with associated memory interfaces. The `rom_intf_req_o` and `rom_intf_rsp_i` signals connect to boot ROM containing the SMC firmware. The ROM must be populated with valid firmware images before initiating the boot sequence.

Scratch RAM interfaces (`scratch_ram_intf_req_o` and `scratch_ram_intf_rsp_i`) connect to 32 independent RAM banks totaling the scratchpad memory space. Cache memory interfaces for L1 instruction and data caches (`l1_icache_*` and `l1_dcache_*`) connect to tag and data RAM banks as specified in the port table.

The `disable_sram_auto_init_i` input controls automatic memory initialization behavior. When deasserted, the SMC automatically zeroes all SRAM contents during bring-up. The `init_mem_done_o` output indicates completion of this initialization sequence. The strap `SRAM_AUTO_ZERO_DISABLE` (GPIO pin 26) can disable this automatic zeroing behavior if your integration requires preserved SRAM contents across resets.

1.1.7. Memory Repair Integration

The SMC implements Built-In Redundancy Analysis (BIRA) memory repair to address manufacturing defects in SRAM

arrays. Repair uses redundant memory elements configured with data stored in eFuse memory, replacing defective cells detected during manufacturing test.

Memory repair executes automatically during boot after eFuse sensing completes and before MBIST operations. The process must complete successfully before the CPU cluster can safely access SRAM arrays. Repair is required on every cold reset but can be bypassed during Function Level Reset (FLR) operations for faster recovery.

Control Signals:

- **fuse_sense_done_o**: eFuse controller output indicating fuse sensing completion. External DFT logic monitors this signal to trigger repair operations.
- **skip_mem_repair_o**: Reset unit output indicating repair should be bypassed. Asserts during FLR operations or when the **BYPASS_SRAM_REPAIR** strap is set. When asserted, external repair logic should skip operations and immediately assert completion status.
- **ext_boot_seq_done_i**: External signal indicating the external boot sequence (memory repair and shadow register override) is complete. Held low to pause reset release until the external agent gives the go-ahead; required before the system can continue its boot sequence

Repair Data:

Repair data is stored in the eFuse BIRA shadow register as a 16-kilobit **repair_data** field. The eFuse controller reads this data during fuse sensing and makes it available through shadow registers. The external memory repair controller must read repair data from **shadow_regs_o** and apply it to SRAM arrays according to your memory architecture.

Status Monitoring:

Repair completion and success are reported through DFT Control Status registers at **0xC000F800** (SOC) and **0xC000F80C** (SEP/SMC). The **mem_repair_done** bit indicates completion regardless of success, while **mem_repair_success** indicates successful repair. Both bits are set by the external repair controller. The SMC ROM firmware monitors these bits during initialization to verify repair completion.

Bypass Configuration:

The **BYPASS_SRAM_REPAIR** configuration strap (GPIO pin 13) bypasses repair logic when asserted high, causing **skip_mem_repair_o** to assert. Use bypass mode only when SRAM arrays are guaranteed defect-free, such as during development or with known-good components.

Integration Requirements:

Provide an external memory repair controller that monitors **fuse_sense_done_o** to initiate repair, respects **skip_mem_repair_o** for bypass control, reads repair data from eFuse shadow registers, applies repair configuration to SRAM arrays, asserts **ext_boot_seq_done_i** and updates DFT Control Status registers with completion status. The controller operates independently of the SMC core and must complete before the CPU cluster accesses SRAM. The testbench demonstrates example integration patterns adaptable to your DFT infrastructure.

For direct access to the eFuse besides from the shim, i.e. from a DFT fuse controller, adopters must ensure the integrated logic will not interfere with accesses from the OCAH's efuse interface.

During FLR operations, repair is automatically bypassed via **skip_mem_repair_o** to enable rapid reset sequences. Normal cold reset sequences always execute repair unless explicitly bypassed via strap configuration.

1.1.8. Interrupt Integration

The SMC aggregates interrupts from multiple sources into `cpu_interrupts_o`, which feeds the PLIC of the embedded RISC-V CPU cluster. The interrupt vector is assembled in `hw/smc/smc_base.sv` and `hw/smc/smc_peripherals/rtl/smc_peripherals.sv`. The authoritative bit-level map (1-core and 4-core columns, RTL trace, and PLIC source ID note) is maintained in a single place:

[SMC CPU interrupt vector map](#)

Configure your interrupt controller to monitor the `sync_irq_o` signal and read SMC interrupt status registers to determine active sources. The testbench demonstrates interrupt injection and handling patterns through the various interrupt input ports.

1.1.9. Telemetry and Debug Integration

For system-level debug and telemetry, the SMC implements ATB (Advanced Trace Bus) receiver interfaces. The telemetry interface group (`telemetry_atdata_i`, `telemetry_atid_i`, `telemetry_atvalid_i`, `telemetry_atready_o`) operates in a dedicated `clk_telemetry_i` clock domain with separate `rst_telemetry_ni` reset. Connect these signals to ATB trace sources for debug data collection.

The `ext_debug_bus_i` provides a 512-bit wide debug observation bus for internal signal monitoring. The cross-trigger interface (`xtrigger_pads_i`, `xtrigger_pads_o`, `xtrigger_ss_i`, `xtrigger_ss_o`) enables coordinated debug triggering across multiple subsystems. The `halt_clock_global_i` input array accepts clock halt requests from external debug agents.

Trace memory interfaces (`trace_mem_req_o` and `trace_mem_resp_i`) connect to trace RAM instances for storing captured debug data. The testbench shows example debug infrastructure connections that can be adapted to your debug requirements.

1.1.10. Bring-Up Sequence

The recommended bring-up sequence begins with asserting `rst_cold_ni` while stabilizing power supplies and clocks. Once all supplies reach nominal voltages, deassert `rst_cold_ni` to initiate the SMC boot sequence. The SMC immediately begins fuse sensing to read configuration values and security credentials.

Monitor the `fuse_sense_done_o` signal to detect completion of fuse sensing. Configuration straps sampled during this phase determine boot modes and operational settings. After fuse sensing completes, the SMC releases internal subsystem resets based on the programmed reset sequencing configuration.

The CPU cluster begins firmware execution from the boot ROM after reset release. The firmware performs additional initialization including peripheral configuration, memory testing (unless disabled), and system health checks. The `init_mem_done_o` signal indicates completion of memory initialization if automatic SRAM zeroing is enabled.

For Function Level Reset (FLR) support, assert `cfg_flr_pf_active_i` to trigger a warm reset sequence. This isolates subsystems through `isolate_req_o` signals and performs controlled reset propagation. Monitor `ss_reset_complete_i` signals to verify subsystem reset completion before releasing isolation.

The testbench file demonstrates the complete power-on sequence including clock initialization, reset timing, strap configuration, and peripheral bring-up. Reference this testbench when implementing your integration-specific bring-up procedures.

1.1.11. SMC Port Interface

The SMC exposes a comprehensive set of interfaces for system integration. The following table defines all SMC top-level ports including clocks, resets, AXI interfaces, peripheral connections, and control signals. Refer to the detailed SMC documentation for internal signal descriptions and timing requirements.

Table 3. SMC Port Declaration

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
clk_smc_i	logic	Input	1	Primary SMC clock domain. All SMC internal logic operates synchronously to this clock.	PLL output	N/A
clk_ref_i	logic	Input	1	Reference clock domain. Used for timing-sensitive operations and peripheral interfaces.	PLL reference clock	N/A
clk_periph_i	logic	Input	1	Peripheral clock domain.	PLL peripheral clock	N/A
powergood_i	logic	Input	1	Power-good indication. Assert after supply voltages stabilize.	Power management controller	N/A
powergood_stable_o	logic	Output	1	Stabilized power-good output. Debounced version of <code>powergood_i</code> .	Reset unit → External systems	N/A
rst_cold_ni	logic	Input	1	Cold reset input (active low). Asynchronous assertion, synchronous deassertion.	System cold reset	N/A
rst_primary_ref_clk_no	logic	Output	1	Primary reset synchronized to reference clock domain (active low).	To subsystems using ref clock	N/A
rst_primary_smc_clk_no	logic	Output	1	Primary reset synchronized to SMC clock domain (active low).	To subsystems using SMC clock	N/A
sys_axi_in_req_i	smc_sys_in_56_64_6_12_axi_req_t	Input	-	System AXI input request (56-bit addr, 64-bit data, 6-bit ID, 12-bit user).	External system masters → Input Fabric	N/A, tie to '0 if unused
sys_axi_in_resp_o	smc_sys_in_56_64_6_12_axi_resp_t	Output	-	System AXI input response.	Input Fabric → External system masters	N/A, leave floating if unused
jtag_axi_in_req_i	smc_jtag_56_64_2_12_axi_req_t	Input	-	JTAG debug AXI request (56-bit addr, 64-bit data, 2-bit ID, 12-bit user).	JTAG debug port → Input Fabric	N/A, tie to '0 if unused
jtag_axi_in_resp_o	smc_jtag_56_64_2_12_axi_resp_t	Output	-	JTAG debug AXI response.	Input Fabric → JTAG debug port	N/A, leave floating if unused
axil_smc_otp_jtag_req_i	smc_pkg::smc_axil_32_32_req_t	Input	-	eFuse OTP/JTAG AXI-Lite request.	JTAG → eFuse controller	N/A, tie to '0 if unused
axil_smc_otp_jtag_resp_o	smc_pkg::smc_axil_32_32_resp_t	Output	-	eFuse OTP/JTAG AXI-Lite response.	eFuse controller → JTAG	N/A, leave floating if unused
sep_axi_in_req_i	smc_sep_in_56_64_6_12_axi_req_t	Input	-	SEP AXI request (56-bit addr, 64-bit data, 6-bit ID, 12-bit user).	SEP → Input Fabric	N/A, tie to '0 if unused
sep_axi_in_resp_o	smc_sep_in_56_64_6_12_axi_resp_t	Output	-	SEP AXI response.	Input Fabric → SEP	N/A, leave floating if unused
output_axi_req_o	smc_sys_out_56_64_8_12_axi_req_t	Output	-	Filtered and remapped AXI output request (56-bit addr, 64-bit data, 8-bit ID, 12-bit user).	Output Fabric → External system resources	N/A, leave floating if unused
output_axi_resp_i	smc_sys_out_56_64_8_12_axi_resp_t	Input	-	Filtered and remapped AXI output response.	External system resources → Output Fabric	N/A, tie to '0 if unused
axil_pll_req_o	smc_axil_32_32_req_t	Output	-	PLL control AXI-Lite request (32-bit addr, 32-bit data).	SMC Fabric → PLL Shim	N/A
axil_pll_resp_i	smc_axil_32_32_resp_t	Input	-	PLL control AXI-Lite response.	PLL Shim → SMC Fabric	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
axil_pvt_req_o	smc_axil_32_32_req_t	Output	-	PVT monitoring AXI-Lite request (32-bit addr, 32-bit data).	SMC Fabric → PVT Shim	N/A
axil_pvt_resp_i	smc_axil_32_32_resp_t	Input	-	PVT monitoring AXI-Lite response.	PVT Shim → SMC Fabric	N/A
efuse_bank_ctrl_req_o	smc_pkg::smc_axil_32_32_req_t	Output	-	eFuse bank control AXI-Lite request.	eFuse controller → eFuse Shim	N/A
efuse_bank_ctrl_resp_i	smc_pkg::smc_axil_32_32_resp_t	Input	-	eFuse bank control AXI-Lite response.	eFuse Shim → eFuse controller	N/A
efuse_shim_command_req_o	smc_efuse_pkg::fuse_command_req_t	Output	-	eFuse command interface request.	eFuse controller → eFuse Shim	N/A
efuse_shim_command_resp_i	smc_efuse_pkg::fuse_command_resp_t	Input	-	eFuse command interface response.	eFuse Shim → eFuse controller	N/A
shadow_regs_o	smc_efuse_pkg::efuse_map_t	Output	-	eFuse shadow register output.	eFuse controller → External systems	N/A
pll_clk_obs_i	logic	Input	1	PLL clock observation input.	PLL → GPIO padding (muxed)	N/A
pll_clk_obs_en_i	logic	Input	1	PLL clock observation enable.	Control logic → GPIO padding	N/A
pvt_process_clk_obs_i	logic	Input	1	PVT process monitor clock observation input.	PVT → GPIO padding (muxed)	N/A
pvt_process_clk_obs_en_i	logic	Input	1	PVT process monitor clock observation enable.	Control logic → GPIO padding	N/A
cat_therm_i	logic	Input	1	CAT thermal sensor input.	CAT thermal sensor → PVT wrap	N/A, tie to '0 if unused
axil_req_gpio_ctrl_o	gpio_pkg::gpio_axil_req_t	Output	-	GPIO controller AXI-Lite request.	SMC Fabric → GPIO padding	N/A
axil_resp_gpio_ctrl_i	gpio_pkg::gpio_axil_resp_t	Input	-	GPIO controller AXI-Lite response.	GPIO padding → SMC Fabric	N/A
lsio_interface_select_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	LSIO interface selection for each GPIO wrap.	GPIO padding → GPIO wraps	N/A
gpio_in_i	logic	Input	[NUM_GPIO_WRAPS-1:0]	GPIO input data from external pins.	External GPIO pins → GPIO padding	N/A
gpio_out_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	GPIO output data to external pins.	GPIO padding → External GPIO pins	N/A
gpio_in_en_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	GPIO input enable control.	GPIO padding → GPIO wraps	N/A
gpio_out_en_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	GPIO output enable control.	GPIO padding → GPIO wraps	N/A
rst_cool_n_from_pin_i	logic	Input	1	Cool reset input from GPIO pin 61.	GPIO pin 61 → Reset Unit	N/A
clk_telemetry_i	logic	Input	1	Telemetry clock domain. Separate clock for ATB telemetry data collection.	Telemetry clock source	N/A
rst_telemetry_n_i	logic	Input	1	Telemetry reset (active low). Resets telemetry receiver logic.	Telemetry reset source	N/A
telemetry_atdata_i	telemetry_receiver_pkg::telemetry_data_t	Input	[NUM_TELMETRY_RECEIVERS-1:0]	ATB telemetry data input array.	ATB sources → Telemetry receivers	N/A
telemetry_atid_i	telemetry_receiver_pkg::atb_id_t	Input	[NUM_TELMETRY_RECEIVERS-1:0]	ATB telemetry ID input array.	ATB sources → Telemetry receivers	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
telemetry_atready_o	logic	Output	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry ready output.	Telemetry receivers → ATB sources	N/A
telemetry_atvalid_i	logic	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry valid input.	ATB sources → Telemetry receivers	N/A
telemetry_atflushvalid_o	logic	Output	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB flush valid output.	Telemetry receivers → ATB sources	N/A
telemetry_atflushready_i	logic	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB flush ready input.	ATB sources → Telemetry receivers	N/A
cluster_double_error_o	logic	Output	1	Cluster double error detection output.	CPU cluster → External systems	N/A
wdt_first_timeout_o	logic	Output	1	First watchdog timeout signal.	CPU cluster WDT → Reset Unit	N/A
wdt_second_timeout_o	logic	Output	1	Second watchdog timeout signal.	Reset Unit → External systems	N/A
smc_region_size_o	logic	Output	[31:0]	SMC address region size output.	SMC Internal Registers → External systems	N/A
ext_interrupts_i	logic	Input	[NUM_EXT_INTERRUPTS-1:0]	External interrupt inputs. Width is config-dependent: 32 (1-core) or 256 (4-core). Maps directly to <code>cpu_interrupts_o[NUM_EXT_INTERRUPTS-1:0]</code> .	External sources → Interrupt aggregator	N/A, tie to '0 if unused
sep_mailbox_interrupts_i	logic	Input	[7:0]	SEP → SMC mailbox interrupt inputs (8 channels). Routed through <code>peripheral_interrupts[7:0] → cpu_interrupts_o[NUM_EXT_INTERRUPTS+7 : NUM_EXT_INTERRUPTS]</code> (1-core: bits 39–32, 4-core: bits 263–256).	SEP → Interrupt aggregator	N/A, tie to '0 if unused
sep_wdt_reset_n_i	logic	Input	1	SEP watchdog reset (active low). Active-low polarity is inverted before routing to <code>peripheral_interrupts[26] → cpu_interrupts_o[NUM_EXT_INTERRUPTS+26]</code> (1-core: bit 58, 4-core: bit 282).	SEP → Interrupt aggregator	N/A
fuse_sense_done_o	logic	Output	1	Fuse sense completion output.	eFuse controller → Memory repair logic	N/A
fuse_reset_n_delayed_o	logic	Output	1	Delayed fuse reset output (after 16 pipeline stages).	eFuse controller → External systems	N/A
skip_mem_repair_o	logic	Output	1	Skip memory repair output.	Reset Unit → Memory repair logic	N/A
ext_boot_seq_done_i	logic	Input	1	External boot sequence completion input (memory repair and shadow register override done); gates reset release.	External boot/repair controller → eFuse controller	N/A
temp_interrupt_i	logic	Input	1	PVT temperature sensor interrupt. Routes to <code>peripheral_interrupts[27] → cpu_interrupts_o[NUM_EXT_INTERRUPTS+27]</code> (1-core: bit 59, 4-core: bit 283).	PVT wrap → Interrupt aggregator	N/A, tie to '0 if unused
lc_state_i	logic	Input	[2*LC_STATE_WIDTH-1:0] (8 bits)	Lifecycle state input, differentially encoded.	Lifecycle controller → SMC Internal Registers	From SEP; tie to 8'hf0 if unused (encoded TEST_DEV)
feat_ctrl_i	sep_lcc_pkg::feat_ctrl_t	Input	-	Feature control vector from lifecycle controller.	SEP lifecycle controller → SMC	From SEP; tie to '0 if unused

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
ras_bank_chip_o	logic	Output	[3:0]	RAS bank chip selection output.	SMC Internal Registers → RAS system	N/A
ras_bank_instance_o	logic	Output	[3:0]	RAS bank instance selection output.	SMC Internal Registers → RAS system	N/A
ext_mailbox_interrupts_o	logic	Output	[NUM_MAILBOXES-1:0]	External mailbox interrupt outputs (32 interrupts, mapped to IDs 156-187).	Mailbox controller → External targets	N/A
cfg_flr_pf_active_i	logic	Input	1	Function Level Reset (FLR) PF active signal.	PCIe/System → Reset Unit	N/A, tie to '0 if unused
isolate_req_o	logic	Output	[31:0]	Isolation request output for 32 subsystems.	Reset Unit → 32 subsystems	N/A
ss_reset_complete_i	logic	Input	[31:0]	Subsystem reset completion status.	32 subsystems → Reset Unit	32'hFFFFFFFF
ss_config_o	logic	Output	[31:0]	Subsystem configuration output.	Reset Unit → 32 subsystems	N/A
ss_reset_ctrl_o	smc_reset_unit_pkg::reset_ctrl_t	Output	[31:0]	Subsystem reset control output array.	Reset Unit → 32 subsystems	N/A
sync_irq_o	logic	Output	1	Synchronized interrupt output.	Interrupt aggregator → External systems	N/A
gpio_interrupt_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	Raw GPIO interrupt vector from the padding, one bit per GPIO wrap (bonded + unbonded). Software reads GPIO status registers to identify the interrupt source. Already OR-reduced into <code>peripheral_interrupts_o[30:29]</code> internally.	<code>smc_padding</code> → external consumers	N/A, leave floating if unused
uart_interrupt_o	logic	Output	[NUM_UART-1:0]	Raw UART interrupt outputs, one bit per UART instance, in the peripheral clock domain. Already aggregated into <code>peripheral_interrupts_o[21:18]</code> internally via CDC.	<code>uart_wrap</code> → external consumers	N/A, leave floating if unused
rom_intf_req_o	rom_req_t	Output	-	ROM memory interface request. Width depends on CPU cluster configuration	CPU cluster → Boot ROM	N/A
rom_intf_rsp_i	rom_rsp_t	Input	-	ROM memory interface response. Width depends on CPU cluster configuration	Boot ROM → CPU cluster	N/A
scratch_ram_intf_req_o	scratch_ram_req_t	Output	[NUM_SRAM_BANKS2-1:0]	Scratch RAM interface requests. Width depends on CPU cluster configuration	CPU cluster → Scratch RAM banks	N/A
scratch_ram_intf_rsp_i	scratch_ram_rsp_t	Input	[NUM_SRAM_BANKS2-1:0]	Scratch RAM interface responses. Width depends on CPU cluster configuration	Scratch RAM banks → CPU cluster	N/A
l1_icache_tag_intf_req_o	l1_icache_tag_req_t	Output	[NUM_ICACHE_TAG_BANKS-1:0]	L1 I-cache tag interface requests. Width depends on CPU cluster configuration	CPU cluster → L1 I-cache tag banks	N/A
l1_icache_tag_intf_rsp_i	l1_icache_tag_rsp_t	Input	[NUM_ICACHE_TAG_BANKS-1:0]	L1 I-cache tag interface responses. Width depends on CPU cluster configuration	L1 I-cache tag banks → CPU cluster	N/A
l1_icache_data_intf_req_o	l1_icache_data_req_t	Output	[NUM_ICACHE_DATA_BANKS-1:0]	L1 I-cache data interface requests. Width depends on CPU cluster configuration	CPU cluster → L1 I-cache data banks	N/A
l1_icache_data_intf_rsp_i	l1_icache_data_rsp_t	Input	[NUM_ICACHE_DATA_BANKS-1:0]	L1 I-cache data interface responses. Width depends on CPU cluster configuration	L1 I-cache data banks → CPU cluster	N/A
l1_dcache_tag_intf_req_o	l1_dcache_tag_req_t	Output	[NUM_DCACHE_TAG_BANKS-1:0]	L1 D-cache tag interface requests. Width depends on CPU cluster configuration	CPU cluster → L1 D-cache tag banks	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
l1_dcache_tag_intf_rsp_i	l1_dcache_tag_rsp_t	Input	[NUM_DCACHE_TAG_BANKS-1:0]	L1 D-cache tag interface responses. Width depends on CPU cluster configuration	L1 D-cache tag banks → CPU cluster	N/A
l1_dcache_data_intf_req_o	l1_dcache_data_req_t	Output	[NUM_DCACHE_DATA_BANKS-1:0]	L1 D-cache data interface requests. Width depends on CPU cluster configuration	CPU cluster → L1 D-cache data banks	N/A
l1_dcache_data_intf_rsp_i	l1_dcache_data_rsp_t	Input	[NUM_DCACHE_DATA_BANKS-1:0]	L1 D-cache data interface responses. Width depends on CPU cluster configuration	L1 D-cache data banks → CPU cluster	N/A
disable_sram_auto_init_i	logic	Input	1	Disable SRAM automatic initialization.	Control logic → CPU cluster	1'b0
init_mem_done_o	logic	Output	1	Memory initialization completion signal.	CPU cluster → Control logic	N/A
chiplet_is_primary_i	logic	Input	1	Chiplet is primary signal. Used for System Timer OCTS	GPIO Pading Straps (strap 25) → SMC	N/A
timer_count_o	logic	Output	[63:0]	64-bit system timer count output for Off-Chip Timer Synchronization (OCTS).	System Timer OCTS → External systems	N/A
rst_cold_override_i	logic	Input	1	Override cold reset value (active low).	Reset override control	1'b1
rst_cool_override_i	logic	Input	1	Override cool reset value (active low).	Reset override control	1'b1
rst_core_override_i	logic	Input	1	Override core reset value (active low).	Reset override control	1'b1
rst_cold_override_en_i	logic	Input	1	Enable cold reset override.	Reset override enable	1'b0
rst_cool_override_en_i	logic	Input	1	Enable cool reset override.	Reset override enable	1'b0
rst_core_override_en_i	logic	Input	1	Enable core reset override.	Reset override enable	1'b0
cla_gpio_enable_o	logic	Output	1	CLA GPIO enable output.	DFD controller → CLA	N/A
tdr_dbg_ctrl_clock_stop_en_i	logic	Input	1	DEBUG_CONTROL CLA clock-stop enable.	DTP → DFD	1'b0
tdr_dbg_ctrl_clocks_stopped_by_cla_o	logic	Output	1	CLA halt-clock status under enable; also drives <code>cpu_interrupts_o[NUM_EXT_INTERRUPTS+64]</code> (internal interrupt 0) — 1-core: bit 96, 4-core: bit 320.	DFD → DTP / interrupt	N/A
halt_clock_global_i	logic	Input	[5:0]	Halt clock global input array.	External XTrigger agents → DFD controller	1'b0
halt_clock_global_o	logic	Output	1	Halt clock global output array.	External XTrigger agents → DFD controller	N/A, leave floating if unused
xtrigger_pads_i	smc_pkg::xtrigger_t	Input	[1:0]	XTrigger pad interface input.	External pads → DFD controller	N/A, tie to '0 if unused
xtrigger_pads_o	smc_pkg::xtrigger_t	Output	[1:0]	XTrigger pad interface output.	DFD controller → External pads	N/A, leave floating if unused
xtrigger_ss_i	smc_pkg::xtrigger_t	Input	[5:0]	XTrigger subsystem interface inputs.	External subsystems → DFD controller	N/A, tie to '0 if unused
xtrigger_ss_o	smc_pkg::xtrigger_t	Output	-	XTrigger subsystem interface output.	DFD controller → External subsystems	N/A, leave floating if unused
trace_mem_req_o	dfd_trace_mem_pkg::SinkMemPktIn_s	Output	[TRC_RAM_INSTANCES-1:0]	Trace memory interface requests.	DFD controller → Trace memory	N/A, leave floating if unused

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
trace_mem_respi	dfd_trace_mem_pkg::SinkMemPktOut_s	Input	[TRC_RAM_INSTANCES-1:0]	Trace memory interface responses.	Trace memory → DFD controller	N/A, tie to '0 if unused
ext_debug_bus_i	logic	Input	[511:0]	External debug bus input (512 bits, 16-bit aligned).	External debug sources → Debug bus aggregator	N/A, tie to '0 if unused
test_en_i	logic	Input	1	DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.	Test controller → All modules	1'b0
scan_rst_ni	logic	Input	1	Scan reset (active-low). Bypasses reset synchronizers during DFT.	Test controller → All modules	1'b1

1.1.12. GPIO Pin Requirements

The SMC provides GPIO functionality with multiplexed peripheral functions, configuration straps, and test interfaces. The following table specifies all GPIO pins with their default configurations, pull-up/pull-down requirements, and strap definitions. Note that strap configurations must be established before cold reset deassertion and remain stable during fuse sensing.

Table 4. OCAH GPIO Requirements

Index	Signal	Bump Loc	Function	Recommended Pull	GPIO Notes
0	smc_wrapper.GPIO_PAD[0]	SMC	SPI.DATA[0]	PD	
1	smc_wrapper.GPIO_PAD[1]	SMC	SPI.DATA[1]	PD	
2	smc_wrapper.GPIO_PAD[2]	SMC	SPI.DATA[2]	PD	
3	smc_wrapper.GPIO_PAD[3]	SMC	SPI.DATA[3]	PD	
4	smc_wrapper.GPIO_PAD[4]	SMC	SPI.DATA[4]	PD	
5	smc_wrapper.GPIO_PAD[5]	SMC	SPI.DATA[5]	PD	
6	smc_wrapper.GPIO_PAD[6]	SMC	SPI.DATA[6]	PD	
7	smc_wrapper.GPIO_PAD[7]	SMC	SPI.DATA[7]	PD	
8	smc_wrapper.GPIO_PAD[8]	SMC	SPI.CS	PU	
9	smc_wrapper.GPIO_PAD[9]	SMC	SPI.CLK	PD	
10	smc_wrapper.GPIO_PAD[10]	SMC	SPI.DQS	PD	
11	smc_wrapper.GPIO_PAD[11]	SMC	UART[0].RX	PU	
12	smc_wrapper.GPIO_PAD[12]	SMC	UART[0].TX	PU	
13	smc_wrapper.GPIO_PAD[13]	SMC	UART[0].RTS	PD	

Index	Signal	Bump Loc	Function	Recommended Pull	GPIO Notes
14	smc_wrapper.GPIO_PAD[14]	SMC	UART[0].CTS	PD	
15	smc_wrapper.GPIO_PAD[15]	SMC	UART[1].RX	PU	
16	smc_wrapper.GPIO_PAD[16]	SMC	UART[1].TX	PU	
17	smc_wrapper.GPIO_PAD[17]	SMC	UART[1].RTS	PD	
18	smc_wrapper.GPIO_PAD[18]	SMC	UART[1].CTS	PD	
19	smc_wrapper.GPIO_PAD[19]	SMC	UART[2].RX	PU	
20	smc_wrapper.GPIO_PAD[20]	SMC	UART[2].TX	PU	
21	smc_wrapper.GPIO_PAD[21]	SMC	UART[2].RTS	PD	
22	smc_wrapper.GPIO_PAD[22]	SMC	UART[2].CTS	PD	
23	smc_wrapper.GPIO_PAD[23]	SMC	UART[3].RX	PU	
24	smc_wrapper.GPIO_PAD[24]	SMC	UART[3].TX	PU	
25	smc_wrapper.GPIO_PAD[25]	SMC	UART[3].RTS	PD	
26	smc_wrapper.GPIO_PAD[26]	SMC	UART[3].CTS	PD	
27	smc_wrapper.GPIO_PAD[27]	SMC	I3C[0] SCL	PU	
28	smc_wrapper.GPIO_PAD[28]	SMC	I3C[0] SDA	PU	
29	smc_wrapper.GPIO_PAD[29]	SMC	I3C[2] SCL	PU	
30	smc_wrapper.GPIO_PAD[30]	SMC	I3C[2] SDA	PU	
31	smc_wrapper.GPIO_PAD[31]	SMC	I3C[3] SCL	PU	
32	smc_wrapper.GPIO_PAD[32]	SMC	I3C[3] SDA	PU	
33	smc_wrapper.GPIO_PAD[33]	SMC	I3C[4] SCL	PU	
34	smc_wrapper.GPIO_PAD[34]	SMC	I3C[4] SDA	PU	
35	smc_wrapper.GPIO_PAD[35]	SMC	I3C[5] SCL	PU	
36	smc_wrapper.GPIO_PAD[36]	SMC	I3C[5] SDA	PU	
37	smc_wrapper.GPIO_PAD[37]	SMC	I2C[0] SCL	PU	Every I2C can be programmed to act as I2C, SMBus or PMBus
38	smc_wrapper.GPIO_PAD[38]	SMC	I2C[0] SDA	PU	

Index	Signal	Bump Loc	Function	Recommended Pull	GPIO Notes
39	smc_wrapper.GPIO_PAD[39]	SMC	I2C[0] SMB_A	PU	SMBus Alert
40	smc_wrapper.GPIO_PAD[40]	SMC	I2C[0] SMB_D	PU	SMBus Suspend
41	smc_wrapper.GPIO_PAD[41]	SMC	I2C[1] SCL	PU	
42	smc_wrapper.GPIO_PAD[42]	SMC	I2C[1] SDA	PU	
43	smc_wrapper.GPIO_PAD[43]	SMC	I2C[1] SMB_A	PU	SMBus Alert
44	smc_wrapper.GPIO_PAD[44]	SMC	I2C[1] SMB_D	PU	SMBus Suspend
45	smc_wrapper.GPIO_PAD[45]	SMC	I2C[2] SCL	PU	
46	smc_wrapper.GPIO_PAD[46]	SMC	I2C[2] SDA	PU	
47	smc_wrapper.GPIO_PAD[47]	SMC	I2C[2] SMB_A	PU	SMBus Alert
48	smc_wrapper.GPIO_PAD[48]	SMC	I2C[2] SMB_D	PU	SMBus Suspend
49	smc_wrapper.GPIO_PAD[49]	SMC	AVS.CLOCK	PU	
50	smc_wrapper.GPIO_PAD[50]	SMC	AVS.MDATA	PU	
51	smc_wrapper.GPIO_PAD[51]	SMC	AVS.SDATA	PU	
52	smc_wrapper.GPIO_PAD[52]	SMC	Thermal Emergency N	PU	
53	smc_wrapper.GPIO_PAD[53]	SMC	Isolate Request	PD	
54	smc_wrapper.GPIO_PAD[54]	SMC	SPI DQS Loopback	PD	
55	smc_wrapper.GPIO_PAD[55]	SMC	System Timer Sync Load	PD	
56	smc_wrapper.GPIO_PAD[56]	SMC	System Timer Credit Count	PD	
57	smc_wrapper.GPIO_PAD[57]	SMC	Boot Stall	PD	Stall after fuse sensing to override shadow register values. Boot will resume when this GPIO is released. This value can be overridden by the value specified over JTAG using the boot_stall_jtag_ovrd and boot_stall_jtag_val signals from DTP.
58	smc_wrapper.GPIO_PAD[58]	SMC	OCCP Interface	PD	
59	smc_wrapper.GPIO_PAD[59]	SMC	Reserved	PD	
60	smc_wrapper.GPIO_PAD[60]	SMC	Reserved	PD	
61	smc_wrapper.UNBONDED_GPIO[61]	SMC	Cool Reset In	PU	Not packaged out
62	smc_wrapper.UNBONDED_GPIO[62]	SMC	Cool Reset Out	PU	Not packaged out

Index	Signal	Bump Loc	Function	Recommended Pull	GPIO Notes
63	smc_wrapper.UNBONDED_GPIO[63]	SMC	I3C[1] SCL	PU	Not packaged out
64	smc_wrapper.UNBONDED_GPIO[64]	SMC	I3C[1] SDA	PU	Not packaged out
65	smc_wrapper.PAD_JTAG_IN_TCK	?	PTAP TCK		
66	smc_wrapper.PAD_JTAG_IN_TRSTN	?	PTAP TRST		
67	smc_wrapper.PAD_JTAG_IN_TDI	?	PTAP TDI		
68	smc_wrapper.PAD_JTAG_IN_TMS	?	PTAP TMS		
69	smc_wrapper.PAD_JTAG_IN_TDO	?	PTAP TDO		
70	smc_wrapper.PAD_JTAG_OUT_TCK	?	STAP TCK		
71	smc_wrapper.PAD_JTAG_OUT_TRSTN	?	STAP TRST		
72	smc_wrapper.PAD_JTAG_OUT_TDI	?	STAP TDI		
73	smc_wrapper.PAD_JTAG_OUT_TMS	?	STAP TMS		
74	smc_wrapper.PAD_JTAG_OUT_TDO	?	STAP TDO		
75	smc_wrapper.PAD_REFCLK	SMC	BP_REFCLK_VDD_SYS		
76	smc_wrapper.PAD_RESETN	SMC	BP_RESETN		
77	smc_wrapper.PAD_POWERGOOD	SMC	BP_POWERGOOD		
78		SEP	OTP[0]		
79		SEP	OTP[1]		
80		SEP	OTP[2]		
81		SEP	OTP[3]		
82		SEP	OTP[4]		
83		SEP	OTP[5]		
84		SMC	PVT[0]		
85		SMC	PVT[1]		
86			VDDIO		
87	smc_wrapper.PLL_CLK_OBS_PAD	SMC	PLL clock OBS	PD	PLL clock observation pin
88	smc_wrapper.PVT_CLK_OBS_PAD	SMC	PVT RO	PD	Process sensor RO output
89					

For multi-chiplet systems, the **PRIMARY_CHIPLET** strap (GPIO pin 25) determines whether the chiplet operates as primary or secondary. The primary chiplet's System Timer OCTS generates synchronization pulses (**timer_sync_load_o**, **timer_cnt_credit_o**) that secondary chiplets receive to maintain coherent timestamps across the system.

1.2. Debug & Test Ports (DTP) Integration

1.2.1. Configuration

Before integrating the DTP, run the configuration script to generate required files:

```
cd hw/dtp
python3 generate_dtp.py --num-ctp <N> --num-int-ct <M>
```

Include the DTP in your build using `hw/dtp/config/dtp.f` or Bender target `dtp_rtl`.

1.2.2. Instantiation

```
dtp #(
    // JTAG Configuration
    .JTAG_BSR_ENABLE      (1'b1),
    .JTAG_TMP_ENABLE      (1'b1),
    .JTAG_SMC_DBG_ENABLE  (1'b1),
    .JTAG_SEP_DBG_ENABLE  (1'b1),
    .JTAG_IDCODE_MFR_ID   (11'h15A),
    .JTAG_IDCODE_PART_NUM (16'h1234),
    .JTAG_IDCODE_SI_REV   (4'h1),
    // Cross Trigger Configuration
    .XTRIG_INT_CT_MODE    (10'b0000000000)
) u_dtp (
    .clk_i          (clk),
    .rst_n_i        (rst_n),
    .pwr_on_rst_ni  (pwr_on_rst_n),
    // JTAG, Boundary Scan, Cross Trigger interfaces...
);
```

1.2.3. Clock and Reset

- **TCK Domain:** JTAG logic operates on external Test Clock
- **System Clock** (`clk_i`): Cross trigger logic and JTAG2AXI bridges
- **System Reset** (`rst_n_i`): Resets cross trigger and AXI; does not affect TAP controller
- **Power-On Reset** (`pwr_on_rst_ni`): Asynchronous reset for JTAG state machine

1.2.4. Constraints

- JTAG interface: 10-50 MHz TCK frequency
- Cross trigger signals synchronized to system clock
- DTP remains active during clock stop for debug access

- External CTP signals require pad ring integration with ESD protection

1.2.5. STAP Scan Chain Topology

The DTP secondary scan path chains multiple STAPs on a single PTAP-managed scan string. The fixed order on this string is: PTAP, I/O STAP (if `JTAG_STAP_IO_ENABLE=1`), SMC debug STAP (if `JTAG_SMC_DBG_ENABLE=1`), SEP debug STAP (if `SEP=1` in the SMU), extra STAP[N-1:0], and return to PTAP.

Both the SMC and SEP debug STAP host interfaces are internal to the SMU and are not exported on the SMU boundary. The SMU wires the SMC debug STAP host outputs to the SMC CPU/system JTAG ports (`smc_cpu_jtag_*`) when `JTAG_SMC_DBG_ENABLE=1`; when disabled, the DTP bypasses this STAP in the secondary scan chain and drives the internal host outputs low. The SEP debug STAP connects to the SEP JTAG debug port when `SEP=1`; when `SEP=0`, the SMU ties `sep_stap_tdo_to_dtp = 1'b0` to bypass the SEP STAP return path.

Each STAP host interface is gated by lifecycle feature controls, synchronized to TCK inside the DTP.

STAP	Lifecycle Host Gating
SMC debug STAP	<code>feat_ctrl_i.soc_debug feat_ctrl_i.ap_debug</code>
SEP debug STAP	<code>feat_ctrl_i.sep_debug feat_ctrl_i.soc_debug feat_ctrl_i.ap_debug</code>

1.2.6. DTP Port Interface

The DTP exposes interfaces for JTAG TAP control, boundary scan, multiple STAP host ports, iJTAG scan chains, AXI debug bridges, cross trigger networks, and CTP GPIO pads. The following table defines all DTP top-level ports. Refer to the dedicated DTP documentation in [hw/dtp/doc/](#) for internal design details.

Table 5. DTP Port Declaration

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
<code>clk_i</code>	logic	Input	1	System clock for cross trigger logic and JTAG2AXI bridges.	System clock	N/A
<code>rst_n_i</code>	logic	Input	1	System reset (active low). Resets cross trigger and AXI logic; does not affect the TAP controller.	System reset	N/A
<code>pwr_on_rst_ni</code>	logic	Input	1	Power-on reset for JTAG state machine (active low, asynchronous).	Power-on reset	N/A
<code>feat_ctrl_i</code>	<code>sep_efuse_pkg::sep_efuse_map_lc_disable_reg_t</code>	Input	-	Lifecycle feature control vector gating debug interfaces. Enable-polarity (1 = feature enabled).	Lifecycle controller	N/A
<code>jtag_ptap_client_tap_ctrl_i</code>	<code>jtag_tap_ctrl_t</code>	Input	-	Primary JTAG TAP control (TCK, TMS, TRST_N).	External JTAG pads	N/A
<code>jtag_ptap_client_tdi_i</code>	logic	Input	1	Primary JTAG TDI.	External JTAG pads	N/A
<code>jtag_ptap_client_tdo_o</code>	logic	Output	1	Primary JTAG TDO.	External JTAG pads	N/A
<code>jtag_ptap_client_tdo_oen_o</code>	logic	Output	1	Primary JTAG TDO output enable.	External JTAG pads	N/A
<code>jtag_bsr_host_scan_ctrl_o</code>	<code>jtag_scan_ctrl_t</code>	Output	-	Boundary scan control. Tie scan_in to scan_out if BSR is disabled.	Boundary scan chain	N/A
<code>jtag_bsr_host_scan_in_i</code>	logic	Input	1	Boundary scan data in.	Boundary scan chain	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
jtag_bsr_host_scan_out_o	logic	Output	1	Boundary scan data out.	Boundary scan chain	N/A
jtag_stap_io_host_tap_ctrl_o	jtag_tap_ctrl_t	Output	-	I/O STAP TAP control for chiplet-to-chiplet connectivity.	I/O STAP host	Tie to '0 if JTAG_STAP_IO_ENABLE=0
jtag_stap_io_host_tdi_i	logic	Input	1	I/O STAP TDI.	I/O STAP host	Tie to '0 if unused
jtag_stap_io_host_tdo_o	logic	Output	1	I/O STAP TDO.	I/O STAP host	N/A
jtag_stap_io_host_tdo_oen_o	logic	Output	1	I/O STAP TDO output enable.	I/O STAP host	N/A
jtag_stap_smc_host_tap_ctrl_o	jtag_tap_ctrl_t	Output	-	SMC debug STAP TAP control.	SMC JTAG port	Tie to '0 if JTAG_SMC_DBG_ENABLE=0
jtag_stap_smc_host_tdi_i	logic	Input	1	SMC debug STAP TDI (from SMC TDO).	SMC JTAG port	Tie to '0 if unused
jtag_stap_smc_host_tdo_o	logic	Output	1	SMC debug STAP TDO (to SMC TDI).	SMC JTAG port	N/A
jtag_stap_smc_host_tdo_oen_o	logic	Output	1	SMC debug STAP TDO output enable.	SMC JTAG port	N/A
jtag_stap_sep_host_tap_ctrl_o	jtag_tap_ctrl_t	Output	-	SEP debug STAP TAP control.	SEP JTAG port	Tie to '0 if JTAG_SEP_DBG_ENABLE=0
jtag_stap_sep_host_tdi_i	logic	Input	1	SEP debug STAP TDI (from SEP TDO).	SEP JTAG port	Tie to '0 if unused
jtag_stap_sep_host_tdo_o	logic	Output	1	SEP debug STAP TDO (to SEP TDI).	SEP JTAG port	N/A
jtag_stap_sep_host_tdo_oen_o	logic	Output	1	SEP debug STAP TDO output enable.	SEP JTAG port	N/A
jtag_stap_extra_host_tap_ctrl_o	jtag_tap_ctrl_t	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TAP control array.	Extra STAP hosts	Tie to '0 if JTAG_NUM_EXTRA_STAPS=0
jtag_stap_extra_host_tdi_i	logic	Input	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDI array.	Extra STAP hosts	Tie to '0 if unused
jtag_stap_extra_host_tdo_o	logic	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDO array.	Extra STAP hosts	N/A
jtag_stap_extra_host_tdo_oen_o	logic	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDO output enable array.	Extra STAP hosts	N/A
jtag_stap_host_scan_ctrl_o	jtag_scan_ctrl_t	Output	-	Extended STAP scan control. Tie scan_in to scan_out if unused.	STAP scan chain	N/A
jtag_stap_host_scan_in_i	logic	Input	1	Extended STAP scan data in.	STAP scan chain	N/A
jtag_stap_host_scan_out_o	logic	Output	1	Extended STAP scan data out.	STAP scan chain	N/A
jtag_dfd_host_scan_ctrl_o	jtag_scan_ctrl_t	Output	-	DFD iJTAG scan control. Tie scan_in to scan_out if unused.	DFD iJTAG chain	N/A
jtag_dfd_host_scan_in_i	logic	Input	1	DFD iJTAG scan data in.	DFD iJTAG chain	N/A
jtag_dfd_host_scan_out_o	logic	Output	1	DFD iJTAG scan data out.	DFD iJTAG chain	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
jtag_dft_secure_host_scan_ctrl_o	jtag_scan_ctrl_t	Output	-	Secure DFT iJTAG scan control. Tie scan_in to scan_out if unused.	Secure DFT iJTAG chain	N/A
jtag_dft_secure_host_scan_in_i	logic	Input	1	Secure DFT iJTAG scan data in.	Secure DFT iJTAG chain	N/A
jtag_dft_secure_host_scan_out_o	logic	Output	1	Secure DFT iJTAG scan data out.	Secure DFT iJTAG chain	N/A
jtag_dft_host_scan_ctrl_o	jtag_scan_ctrl_t	Output	-	Non-secure DFT iJTAG scan control. Tie scan_in to scan_out if unused.	Non-secure DFT iJTAG chain	N/A
jtag_dft_host_scan_in_i	logic	Input	1	Non-secure DFT iJTAG scan data in.	Non-secure DFT iJTAG chain	N/A
jtag_dft_host_scan_out_o	logic	Output	1	Non-secure DFT iJTAG scan data out.	Non-secure DFT iJTAG chain	N/A
axi_smc_dbg_req_o	smc_jtag_axi_req_t	Output	-	SMC fabric debug AXI request. Active when JTAG_SMC_DBG_ENABLE=1.	SMC jtag_axi_in_req_i	Tie to '0 if disabled
axi_smc_dbg_resp_i	smc_jtag_axi_resp_t	Input	-	SMC fabric debug AXI response.	SMC jtag_axi_in_resp_o	Tie to '0 if disabled
axil_smc_otp_jtag_req_o	smc_otp_axil_req_t	Output	-	SMC OTP debug AXI-Lite request.	SMC axil_smc_otp_jtag_req_i	Tie to '0 if disabled
axil_smc_otp_jtag_resp_i	smc_otp_axil_resp_t	Input	-	SMC OTP debug AXI-Lite response.	SMC axil_smc_otp_jtag_resp_o	Tie to '0 if disabled
axil_sep_otp_jtag_req_o	sep_otp_axil_req_t	Output	-	SEP OTP debug AXI-Lite request.	SEP axil_sep_otp_jtag_req_i	Tie to '0 if disabled
axil_sep_otp_jtag_resp_i	sep_otp_axil_resp_t	Input	-	SEP OTP debug AXI-Lite response.	SEP axil_sep_otp_jtag_resp_o	Tie to '0 if disabled
stop_clks_o	logic	Output	1	Clock stop output from JTAG DEBUG_CONTROL or CLA clock-stop requests via CTN.	PLL clock gates	N/A
cla_clock_stop_en_o	logic	Output	1	CLA clock stop enable from DEBUG_CONTROL register.	Clock control logic	N/A
jtag_boot_stall_ovrd_o	logic	Output	1	JTAG boot stall override enable.	Boot control	N/A
jtag_boot_stall_o	logic	Output	1	JTAG boot stall value when override is asserted.	Boot control	N/A
jtag_ic_reset_smc_o	ic_reset_smc_t	Output	\$bits(ic_reset_smc_t)	SMC IC_RESET override slice. Struct contains .ovrd (override enables) and .val (active-low override values), each sized to match the corresponding SMC reset control struct. Tied to '0 when JTAG_IC_RESET_SMC_ENABLE=0.	SMC jtag_smc_reset_ctrl_i	N/A
jtag_ic_reset_sep_o	ic_reset_sep_t	Output	\$bits(ic_reset_sep_t)	SEP IC_RESET override slice. Struct contains .ovrd (override enables) and .val (active-low override values), each sized to match the corresponding SEP reset control struct. Tied to '0 when JTAG_IC_RESET_SEP_ENABLE=0.	SEP jtag_sep_reset_ctrl_i	N/A
jtag_ic_reset_ext_o	ic_reset_ext_t	Output	\$bits(ic_reset_ext_t)	External IC_RESET override slice. Struct contains .ovrd (override enables) and .val (active-low override values), each sized per integrator-defined type. Tied to '0 when JTAG_IC_RESET_EXT_ENABLE=0.	Integrator-defined reset overrides	N/A
jtag_ptap_state_o	tap_state_e	Output	-	Current PTAP TAP state.	Debug infrastructure	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
jtag_ptap_inst_decoded_o	jtag_instruction_decoded_e	Output	-	Current decoded PTAP instruction.	Debug infrastructure	N/A
axil_xtrig_req_i	xtrig_axil_req_t	Input	-	Cross trigger CSR AXI-Lite request.	SMC axil_dtp_csr_req_o	N/A
axil_xtrig_resp_o	xtrig_axil_resp_t	Output	-	Cross trigger CSR AXI-Lite response.	SMC axil_dtp_csr_resp_i	N/A
xtrig_ctm_src_req_o	logic	Output	[XTRIG_NUM_INT_CT-1:0]	Cross trigger source requests from CTM to destination sinks.	Cross trigger network	N/A
xtrig_ctm_src_ack_i	logic	Input	[XTRIG_NUM_INT_CT-1:0]	Cross trigger source acknowledges. Unused in pulse-sync mode (mode bit = 0).	Cross trigger network	Tie to '0 in pulse-sync mode
xtrig_ctm_dst_req_i	logic	Input	[XTRIG_NUM_INT_CT-1:0]	Cross trigger destination requests from source generators.	Cross trigger network	Tie to '0 if unused
xtrig_ctm_dst_ack_o	logic	Output	[XTRIG_NUM_INT_CT-1:0]	Cross trigger destination acknowledges. Unused in pulse-sync mode (mode bit = 0).	Cross trigger network	N/A
xtrig_clk_stop_req_i	logic	Input	[XTRIG_NUM_CLK_STOP_REQ-1:0]	Clock stop requests from CLAs or other devices in the chiplet.	Clock stop requestors	Tie to '0 if unused
xtrig_ctp_req_out_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-out data output (to GPIO pad ring).	GPIO pad ring	N/A
xtrig_ctp_req_out_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-out data output enable.	GPIO pad ring	N/A
xtrig_ctp_req_out_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP request-out data input (from GPIO pad ring).	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_req_out_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-out data input enable.	GPIO pad ring	N/A
xtrig_ctp_req_in_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data output.	GPIO pad ring	N/A
xtrig_ctp_req_in_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data output enable.	GPIO pad ring	N/A
xtrig_ctp_req_in_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP request-in data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_req_in_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data input enable.	GPIO pad ring	N/A
xtrig_ctp_ack_in_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data output.	GPIO pad ring	N/A
xtrig_ctp_ack_in_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data output enable.	GPIO pad ring	N/A
xtrig_ctp_ack_in_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP ack-in data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_ack_in_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data input enable.	GPIO pad ring	N/A
xtrig_ctp_ack_out_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data output.	GPIO pad ring	N/A
xtrig_ctp_ack_out_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data output enable.	GPIO pad ring	N/A
xtrig_ctp_ack_out_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP ack-out data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_ack_out_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data input enable.	GPIO pad ring	N/A

1.3. Security Processor (SEP) Integration

The Security Processor (SEP) is the Root of Trust for the chiplet, providing a physically isolated execution environment for security-critical services such as secure boot, lifecycle control, device attestation, cryptographic acceleration, and key management. This section provides practical integration procedures for adopters integrating the SEP as part of the SMU or as a standalone out-of-context IP.

For detailed architectural documentation about the SEP internal design, refer to the dedicated SEP documentation located in [hw/sep/doc/](#). The SEP hardware overview and interconnect fabric are described in [hw/sep/doc/overview.adoc](#) and [hw/sep/doc/fabric.adoc](#) respectively.

1.3.1. Module Overview

The `sep.sv` module provides the complete SEP functionality with logic-level interfaces. Technology-specific components (SRAM, ROM, eFuse shims, external TRNG, SPI controller selection) are instantiated outside SEP in a `sep_ip_integration` block, following the same vendor-neutral philosophy used for the SMC. When integrated inside the SMU, the SEP's external interfaces are passed through the SMU boundary with a `sep_` prefix.

1.3.2. Clock and Reset Requirements

The SEP requires two clock inputs. Connect `clk_i` to the primary system clock driving all SEP internal logic. Connect `clk_wdt_i` to a 200 kHz clock source for the watchdog timer; this clock must be independent of the system clock so that the WDT can detect system clock failures.

The primary reset `rst_ni` follows the standard active-low, asynchronous assertion / synchronous deassertion convention. The `dbg_rstb_i` input provides a separate reset for the EL2 CPU debug module. The `wdt_rst_ni` input receives the watchdog reset generated by the SMC reset unit. The SEP outputs `sep_reset_n_o` for its IP integration block and `wdt_timer_rst_req_o` when the WDT triggers a reset request.

1.3.3. AXI Interfaces

The SEP connects to the system through three AXI paths. The `smn_outbound_axi` master interface allows SEP firmware to access system resources outside the SEP domain, such as SMC registers or external memory. The `smn_inbound_axi` slave interface accepts requests from system masters requiring access to SEP-managed resources. The `sep_ext_to_smc_axi` output provides a dedicated AXI path for SEP-to-SMC transactions, connecting to the SMC's `sep_axi_in` port.

An additional AXI extension port (`axi_extension_axi`) enables adopters to attach custom peripherals to the SEP interconnect fabric without modifying the SEP core. When unused, tie the response input to return DECERR.

1.3.4. Memory Interfaces

All SEP memory macros are instantiated externally and connected through dedicated request/response interfaces. The `sep_cpu_tcm` interface provides tightly-coupled memory (ICCM and DCCM) for the VeeR EL2 CPU. The `sep_sram` interface connects scratch SRAM used for firmware data. The `sep_boot_rom` interface provides the BL0 ROM containing the first-stage bootloader.

The crypto subsystem requires additional memory interfaces. `sep_crypto_pka_imem_sram` and `sep_crypto_pka_dmem_sram` connect to the OTBN public key accelerator instruction and data memories respectively. `km_rom_mem` and `km_sram_mem` connect to the Key Manager's dedicated ROM and SRAM macros.

1.3.5. Cryptographic Subsystem Connections

The SEP includes an internal entropy source that requires a ring-oscillator sample clock connected to `entropy_rosc_sample_clk_i`. This clock must be asynchronous to `clk_i` and typically runs between 100 and 400 MHz.

For adopters providing an external TRNG, the `ext_trng_axil` AXI-Lite interface connects to the external TRNG control registers (in `sep_ip_integration`), while the `ext_trng_axis` AXI-Stream interfaces carry entropy data. The `ext_trng_irq_i` input is routed to the PIC; `ext_trng_alarm_i` is reserved for RAS.

1.3.6. eFuse and Lifecycle

The SEP accesses its own eFuse bank through the `efuse_bank_ctrl` AXI-Lite interface and the `efuse_shim_command` custom interface, mirroring the pattern used by the SMC. The SEP must wait for the SMC fuse sense to complete (`smc_fuse_sense_done_i`) before initiating its own fuse operations, and it signals completion through `sep_fuse_sense_done_o`.

The lifecycle controller outputs `lc_state_o` (differentially encoded), `feat_ctrl_o` (feature control vector), `lc_sigint_err_o` (signal integrity error), and `security_disable_o`. These outputs gate debug access through the DTP and control feature availability across the chiplet. The `lcc_demote_state` outputs report lifecycle demotion events to the SMC.

An OTP debug AXI-Lite port (`axil_sep_otp_jtag`) connects to the DTP for JTAG access to SEP fuses. When SEP is disabled (`SEP=0` at the SMU level), the SMU instantiates an AXI-Lite error slave on this path.

1.3.7. SPI Host Interface

The SEP provides a SPI host controller through the `sep_io_spi` interface, supporting single, quad, and octal modes. The `spi_irq_i` input receives the muxed SPI interrupt from `sep_ip_integration`, which selects between the internal OpenTitan SPI controller and an external SPI controller. The `ot_spi_irq_o` output (at the SMU level) carries the OpenTitan SPI interrupt for external muxing.

1.3.8. Bring-Up Considerations

The SEP boot sequence begins after the SMC completes fuse sensing and releases the SEP reset. The SEP BL0 ROM firmware verifies the BL1 software image using non-volatile public keys before transferring control. Feature controls derived from secure boot authentication and lifecycle state govern what operations are permitted after boot.

The `sep_straps_i` input provides configuration straps for SEP-specific settings. The `rst_vec`, `jtag_id`, and `core_id` inputs must be tied to appropriate constants in the top-level integration.

1.3.9. SEP Port Interface

The SEP exposes interfaces for clocks, resets, JTAG debug, CPU control, memory macros, AXI bus connections, cryptographic subsystem, eFuse, lifecycle, SPI host, and mailbox. The following table defines all SEP top-level ports.

Table 6. SEP Port Declaration

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
<code>clk_i</code>	<code>logic</code>	Input	1	Primary SEP clock. All SEP internal logic operates synchronously to this clock.	System clock	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
clk_wdt_i	logic	Input	1	200 kHz clock for watchdog timer.	Low-frequency clock source	N/A
rst_ni	logic	Input	1	Primary reset (active low). Asynchronous assertion, synchronous deassertion.	System reset	N/A
dbg_rstb_i	logic	Input	1	EL2 debugger reset (active low).	JTAG debug infrastructure	N/A
wdt_rst_ni	logic	Input	1	Watchdog timer reset from SMC (active low).	SMC reset unit	N/A
sep_reset_n_o	logic	Output	1	Reset for SEP IP integration block (active low).	sep_ip_integration	N/A
wdt_timer_rst_req_o	logic	Output	1	WDT reset request output. Triggers memory interface reset in sep_wrapper .	SEP wrapper	N/A
jtag_tck	logic	Input	1	JTAG test clock.	JTAG pads (via DTP STAP)	N/A
jtag_tms	logic	Input	1	JTAG test mode select.	JTAG pads (via DTP STAP)	N/A
jtag_tdi	logic	Input	1	JTAG test data in.	JTAG pads (via DTP STAP)	N/A
jtag_trst_n	logic	Input	1	JTAG test reset (active low).	JTAG pads (via DTP STAP)	N/A
jtag_tdo	logic	Output	1	JTAG test data out.	JTAG pads (via DTP STAP)	N/A
jtag_tdoEn	logic	Output	1	JTAG test data out enable.	JTAG pads (via DTP STAP)	N/A
axil_sep_otp_jtag_req_i	sep_efuse_pkg::efuse_axil_req_t	Input	-	OTP debug AXI-Lite request from JTAG.	DTP axil_sep_otp_jtag_req_o	Tie to '0 if unused
axil_sep_otp_jtag_resp_o	sep_efuse_pkg::efuse_axil_resp_t	Output	-	OTP debug AXI-Lite response to JTAG.	DTP axil_sep_otp_jtag_resp_i	N/A
mpc_debug_halt_req	logic	Input	1	Async halt request from MPC.	Debug controller	Tie to '0 if unused
mpc_debug_run_req	logic	Input	1	Async run request from MPC.	Debug controller	Tie to '0 if unused
mpc_reset_run_req	logic	Input	1	Run/halt selection after reset.	Debug controller	1'b1
i_cpu_halt_req	logic	Input	1	Async halt request to CPU.	Debug controller	Tie to '0 if unused
i_cpu_run_req	logic	Input	1	Async restart request to CPU.	Debug controller	Tie to '0 if unused
test_en_i	logic	Input	1	DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.	Test controller	1'b0
scan_rst_ni	logic	Input	1	Scan reset (active-low). Bypasses reset synchronizers during DFT.	Test controller	1'b1
ext_boot_seq_done_i	logic	Input	1	External boot sequence completion (memory repair and shadow register override done); gates reset release.	External boot/repair controller	1'b1
dmi_core_enable	logic	Input	1	DMI core enable.	Debug infrastructure	Tie to '0 if unused
dmi_uncore_enable	logic	Input	1	DMI uncore enable.	Debug infrastructure	Tie to '0 if unused

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
dmi_uncore_en	logic	Output	1	DMI uncore access enable.	Debug infrastructure	N/A
dmi_uncore_wr_en	logic	Output	1	DMI uncore write enable.	Debug infrastructure	N/A
dmi_uncore_addr	logic	Output	[6:0]	DMI uncore address.	Debug infrastructure	N/A
dmi_uncore_wdata	logic	Output	[31:0]	DMI uncore write data.	Debug infrastructure	N/A
dmi_uncore_rdata	logic	Input	[31:0]	DMI uncore read data.	Debug infrastructure	N/A
dmi_active	logic	Output	1	DMI active indicator.	Debug infrastructure	N/A
sep_cpu_trace	sep_cpu_trace_t	Output	-	CPU trace output for debug.	Trace infrastructure	N/A
jtag_id	logic	Input	[31:1]	JTAG ID value.	Configuration tie-off	N/A
core_id	logic	Input	[31:4]	CPU core ID.	Configuration tie-off	N/A
timer_int	logic	Input	1	Timer interrupt.	Internal timer	Tie to '0 if unused
soft_int	logic	Input	1	Software interrupt.	Software IRQ source	Tie to '0 if unused
extintsrc_req	logic	Input	[NUM_EXTERNAL_IRQS-1:0]	External interrupt sources for PIC.	External IRQ sources	Tie to '0 if unused
sep_cpu_tcm_req_o	sep_cpu_tcm_req_t	Output	-	CPU tightly-coupled memory (TCM) request (ICCM + DCCM).	SRAM macros	N/A
sep_cpu_tcm_rsp_i	sep_cpu_tcm_rsp_t	Input	-	CPU TCM response.	SRAM macros	N/A
sep_sram_req	sep_sram_req_t	Output	-	Scratch SRAM request.	SRAM macros	N/A
sep_sram_rsp	sep_sram_rsp_t	Input	-	Scratch SRAM response.	SRAM macros	N/A
sep_boot_rom_req	sep_sram_req_t	Output	-	Boot ROM (BL0) request.	ROM macro	N/A
sep_boot_rom_rsp	sep_sram_rsp_t	Input	-	Boot ROM (BL0) response.	ROM macro	N/A
smn_outbound_axi_req_o	sep_system_peripherals_outbound_axi_req_t	Output	-	SMN outbound AXI request (system bus master).	SMN fabric	N/A
smn_outbound_axi_resp_i	sep_system_peripherals_outbound_axi_resp_t	Input	-	SMN outbound AXI response.	SMN fabric	N/A
smn_inbound_axi_req_i	sep_system_peripherals_internal_axi_req_t	Input	-	SMN inbound AXI request (system bus slave).	SMN fabric	Tie to '0 if unused
smn_inbound_axi_resp_o	sep_system_peripherals_internal_axi_resp_t	Output	-	SMN inbound AXI response.	SMN fabric	N/A
sep_ext_to_smc_axi_req_o	sep_system_peripherals_internal_axi_req_t	Output	-	SEP-to-SMC AXI request. Dedicated path for SEP to access SMC resources.	SMC sep_axi_in_req_i	N/A
sep_ext_to_smc_axi_resp_i	sep_system_peripherals_internal_axi_resp_t	Input	-	SEP-to-SMC AXI response.	SMC sep_axi_in_resp_o	N/A
entropy_rosc_sample_clk_i	logic	Input	1	Ring-oscillator sample clock for entropy source (asynch vs clk_i; ~100-400 MHz typical).	Ring oscillator	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
sep_crypto_pka_imem_sram_req	sep_crypto_pka_imem_sram_req_t	Output	-	OTBN PKA instruction memory SRAM request.	SRAM macro	N/A
sep_crypto_pka_imem_sram_rsp	sep_crypto_pka_imem_sram_rsp_t	Input	-	OTBN PKA instruction memory SRAM response.	SRAM macro	N/A
sep_crypto_pka_dmem_sram_req	sep_crypto_pka_dmem_sram_req_t	Output	-	OTBN PKA data memory SRAM request.	SRAM macro	N/A
sep_crypto_pka_dmem_sram_rsp	sep_crypto_pka_dmem_sram_rsp_t	Input	-	OTBN PKA data memory SRAM response.	SRAM macro	N/A
ext_trng_axil_req_o	sep_32_32_axil_req_t	Output	-	External TRNG AXI-Lite passthrough request (to sep_ip_integration).	External TRNG shim	N/A
ext_trng_axil_resp_i	sep_32_32_axil_resp_t	Input	-	External TRNG AXI-Lite passthrough response.	External TRNG shim	N/A
ext_trng_axis_req_i	ext_trng_axis_req_t	Input	[EXT_TRNG_NUM_AXIS-1:0]	External TRNG AXI-Stream request array (from sep_ip_integration).	External TRNG shim	N/A
ext_trng_axis_rsp_o	ext_trng_axis_rsp_t	Output	[EXT_TRNG_NUM_AXIS-1:0]	External TRNG AXI-Stream response array.	External TRNG shim	N/A
ext_trng_irq_i	logic	Input	1	External TRNG interrupt (routed to PIC).	External TRNG shim	Tie to '0 if unused
ext_trng_alarm_i	logic	Input	1	External TRNG alarm (reserved for RAS).	External TRNG shim	Tie to '0 if unused
km_rom_mem_req_o	km_intf_pkg::km_rom_mem_req_t	Output	-	Key Manager ROM memory request.	ROM macro	N/A
km_rom_mem_rsp_i	km_intf_pkg::km_rom_mem_rsp_t	Input	-	Key Manager ROM memory response.	ROM macro	N/A
km_sram_mem_req_o	km_intf_pkg::km_sram_mem_req_t	Output	-	Key Manager SRAM memory request.	SRAM macro	N/A
km_sram_mem_rsp_i	km_intf_pkg::km_sram_mem_rsp_t	Input	-	Key Manager SRAM memory response.	SRAM macro	N/A
efuse_bank_ctrl_req_o	sep_efuse_pkg::efuse_axil_req_t	Output	-	eFuse bank control AXI-Lite request.	eFuse shim	N/A
efuse_bank_ctrl_resp_i	sep_efuse_pkg::efuse_axil_resp_t	Input	-	eFuse bank control AXI-Lite response.	eFuse shim	N/A
efuse_shim_command_req_o	sep_efuse_pkg::fuse_command_req_t	Output	-	eFuse command interface request.	eFuse shim	N/A
efuse_shim_command_resp_i	sep_efuse_pkg::fuse_command_resp_t	Input	-	eFuse command interface response.	eFuse shim	N/A
lcc_demote_state_1_o	logic	Output	[1:0]	Lifecycle demote state output 1 (to SMC).	SMC lifecycle logic	N/A
lcc_demote_state_2_o	logic	Output	[1:0]	Lifecycle demote state output 2 (to SMC).	SMC lifecycle logic	N/A
sep_io_spi_req_o	sep_io_spi_req_t	Output	-	SPI host controller request.	SPI pad interface	N/A
sep_io_spi_rsp_i	sep_io_spi_rsp_t	Input	-	SPI host controller response.	SPI pad interface	N/A
spi_irq_i	logic	Input	1	Muxed SPI IRQ from sep_ip_integration .	SPI controller	Tie to '0 if unused
lc_state_o	logic	Output	[2*LC_STATE_BIT_WIDTH-1:0]	Lifecycle state output (differentially encoded).	SMC, DTP	N/A
feat_ctrl_o	sep_efuse_pkg::sep_efuse_map_lc_disable_reg_t	Output	-	Lifecycle feature control vector.	SMC, DTP	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
lc_sigint_err_o	logic	Output	1	Lifecycle signal integrity error.	SMC	N/A
security_disable_o	logic	Output	1	Security disable output.	SMC	N/A
smc_mailbox_interrupt_o	logic	Output	[NUM_MAILBOXES-1:0]	Mailbox interrupt outputs to SMC.	SMC interrupt aggregator	N/A
smc_fuse_sense_done_i	logic	Input	1	SMC fuse sense done indication.	SMC fuse_sense_done_o	N/A
sep_fuse_sense_done_o	logic	Output	1	SEP fuse sense done output.	System integration	N/A
sep_straps_i	sep_pkg::sep_straps_t	Input	-	SEP configuration straps.	GPIO padding / tie-off	N/A
axi_extension_axi_req_o	sep_32_64_6_12_axi_req_t	Output	-	AXI extension port request for adopter peripheral expansion.	Adopter peripherals	N/A
axi_extension_axi_resp_i	sep_32_64_6_12_axi_resp_t	Input	-	AXI extension port response.	Adopter peripherals	Tie to DECERR if unused
smc_global_base_addr_i	logic	Input	[55:0]	SMC global base address for outbound address translation.	System configuration	N/A
smc_region_size_i	logic	Input	[55:0]	SMC region size for outbound address translation.	SMC smc_region_size_o	N/A
km_unrecoverable_err_o	logic	Output	1	Key Manager unrecoverable error.	RAS / error handling	N/A
km_recoverable_err_o	logic	Output	1	Key Manager recoverable error.	RAS / error handling	N/A
ext_debug_bus_o	logic	Output	[511:0]	External debug observation bus.	Debug infrastructure	N/A

1.4. SMU Composition and Internal Wiring

The System Management Unit (SMU) combines the SMC, DTP, and optionally the SEP into a unified subsystem. The interface descriptions in the preceding sections apply directly when integrating the SMU, since the SMU inherits all three subsystems' ports. This section covers the module variants available to adopters, how the SMU wires its subsystems together internally, and what additional configuration it exposes.

1.4.1. Module Variants and IP Integration

OCAH core modules (`smc.sv`, `sep.sv`, `dtp.sv`, and the composite `smu.sv`) expose logic-level interfaces only. All technology-specific components — PLLs, PVT sensors, eFuse shims, memory macros, GPIO padings, SPI controllers, and external TRNG — are instantiated outside the core modules by the adopter. This separation keeps the OCAH deliverable technology-neutral.

The repository provides reference wrappers that demonstrate how to bridge the gap between OCAH core interfaces and adopter-specific IP:

- `smu_wrapper.sv` wraps `smu.sv` with pad-level signal handling for clocks, resets, GPIO, eFuse, and JTAG interfaces. It internally instantiates `smc_ip_integration.sv` for SMC-related vendor IP and (when `SEP=1`) `sep_ip_integration.sv` for SEP-related vendor IP.
- `smc_wrapper.sv` provides an equivalent reference wrapping `smc.sv` alone, useful when integrating the SMC without

the DTP/SEP subsystems.

Neither the wrappers nor the IP integration modules are part of the OCAH deliverable; they exist as integration guides and for Tenstorrent internal usage. Adopters must develop their own equivalents following the interface patterns demonstrated in these modules. The interface contracts at the module boundaries ensure technology substitution without modifying the verified OCAH logic.

The figure below illustrates the relationship between the OCAH core modules, the wrapper layer, and the IP integration modules.

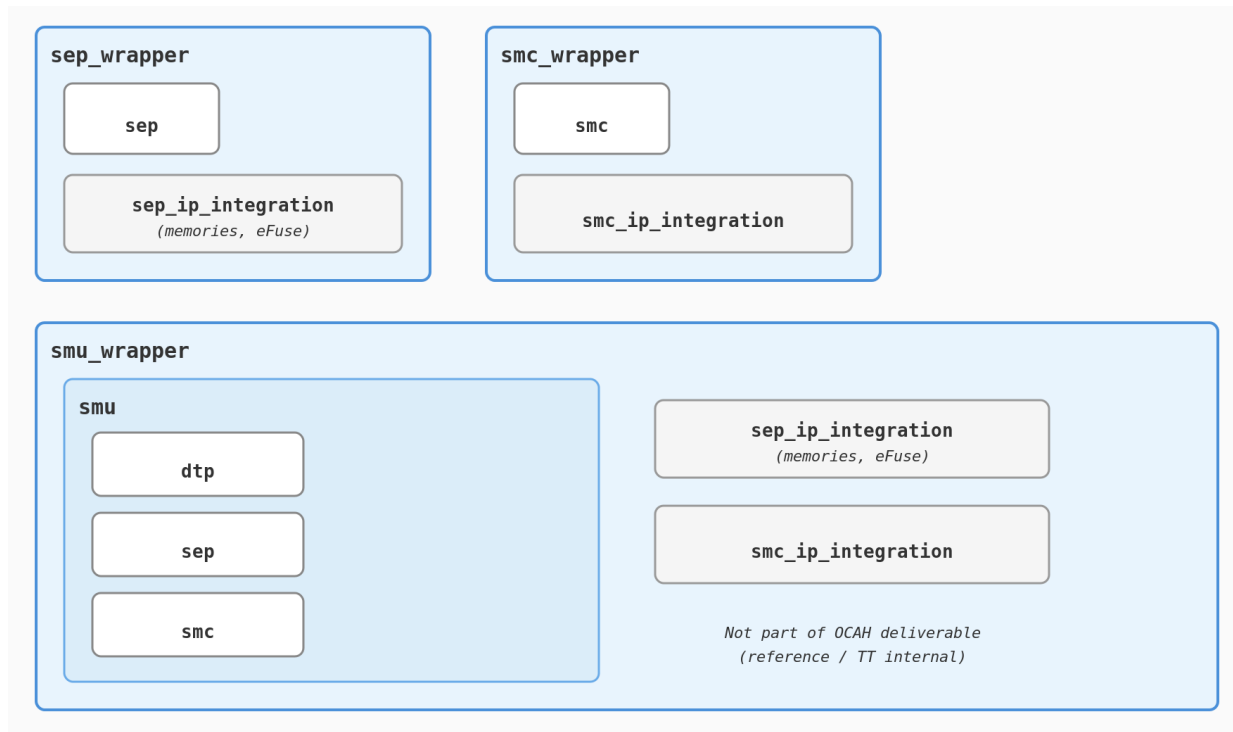


Figure 1. SMU Integration Components

IP Integration Modules

smc_ip_integration.sv demonstrates integration of SMC-side technology components: PLL wrappers (generating **pll_cgm_clk_o** and **pll_awnm_clk_o**), PVT sensor interfaces, eFuse shims, GPIO padding logic, memory macros (ROM, scratchpad RAM, cache tag and data arrays), and trace memory instances.

sep_ip_integration.sv demonstrates integration of SEP-side technology components: SEP memory macros (TCM, scratch SRAM, boot ROM, OTBN IMEM/DMEM, Key Manager ROM/SRAM), eFuse shim, external TRNG, and SPI controller selection.

Wrapper Clock and PLL Architecture

The **smu_wrapper.sv** (and **smc_wrapper.sv**) reference accepts **BP_REFCLK** at the pad level and processes it through the integrated padding to produce internal reference timing. A **pll_wrapper** module generates system clocks **pll_cgm_clk_o** and **pll_awnm_clk_o** from the padding-processed reference. The SMC controls this PLL through the same AXI-Lite interface exposed by the core module. However, the core receives its clocks through DFX inputs **pll_cgm_clk_dfx_i** and **ref_clk_vdd_sys_dfx_i**, so the adopter must route PLL outputs through the clock distribution network and feed them back through these DFX inputs.

1.4.2. Clock and Reset Architecture

Connect `clk_smu_i` to the primary system clock and `clk_ref_i` to a stable reference source. The DTP operates on TCK for JTAG logic and on the system clock for cross-trigger and AXI bridges. Reset sequencing follows the SMC flow; the DTP JTAG state machine uses `pwr_on_rst_ni`, which the SMU connects to `rst_cold_ni`. The SMU internally connects DTP debug paths to the SMC and optionally instantiates the SEP when the `SEP` parameter is set to 1. The full interconnection details are described in [SMC-DTP Internal Interconnection](#) and [SEP-Conditional Wiring](#) below.

1.4.3. SMU AXI Crossbar Address Map

Inside the SMU, the `smu_axi_xbar` module routes traffic between three AXI4 manager/subordinate pairs: the SEP outbound master (`sep_out`), the SMC outbound master (`smc_out`), and the external SMU AXI input (`ext_in` / `smu_axi_in`). The corresponding targets are the SEP and SMC inbound subordinates, and the external SMU AXI output (`ext_out` / `smu_axi_out`). Connectivity is described in [SMU AXI Crossbar](#); the split between hardware-fixed behaviour and firmware-programmable apertures is documented in [OCAH Memory Map](#).

The crossbar uses a **runtime** address map. Only the SEP and SMC apertures are programmable at runtime; the `ext_out` rule is static:

- `sep_global_base_addr_i` / `sep_region_size_i` come from the `sep_cpu_ctrl` CSRs `SEP_GLOBAL_BASE_ADDR` / `SEP_REGION_SIZE`, exposed at both the `sep.sv` and `smu.sv` boundaries as `sep_global_base_addr_o` / `sep_region_size_o` (both 56-bit). `SEP_REGION_SIZE` is a 32-bit CSR field, zero-extended to 56-bit at the SEP system CSR; the xbar input takes only the lower 32 bits. `SEP_REGION_SIZE` defines the SEP global/chiplet aperture for SMU routing and SEP inbound global-to-local remap. It is distinct from the SEP CPU's architectural local-alias span. `SEP_REGION_SIZE` currently resets to `0x0100_0000` (16 MiB). Tied to '0' at the `smu.sv` boundary when the `SEP` parameter is 0.
- `smc_global_base_addr_i` / `smc_region_size_i` come from the SMC CPU control CSRs `GLOBAL_BASE` / `REGION_SIZE`, exposed at both the `smc.sv` and `smu.sv` boundaries as `smc_global_base_o` (56-bit) / `smc_region_size_o` (32-bit). `REGION_SIZE` resets to `0x0100_0000` (16 MiB).

SEP, SMC, and `smu_axi_xbar` all run on `clk_smu_i`, so these quasi-static CSR fields need no clock-domain crossing. The crossbar's address arithmetic is done on 57-bit widened sums; `ext_out` covers `0x0000_8000_0000` up to the top of the 56-bit address space.

Firmware programming order

Firmware must program `SEP_GLOBAL_BASE_ADDR` / `SEP_REGION_SIZE` and SMC `GLOBAL_BASE` / `REGION_SIZE` during early boot, **before any cross-subsystem AXI traffic is issued**, and must ensure the two apertures do not overlap and both fall within the chiplet-visible SMU envelope assigned by the integrator. The SMU crossbar does not itself constrain envelope size; see [OCAH Memory Map](#) for the responsibility split between integrator, firmware, and the internal SEP/SMC address maps.

1.4.4. SMU Default Parameters

The SMU exposes a set of configurable parameters that control DTP JTAG features, identification, cross trigger, and pipeline depths. The table below lists each parameter with its default value.

Parameter	Value	Description
<code>JTAG_BSR_ENABLE</code>	1	Boundary scan enabled
<code>JTAG_EXTTEST_TRAIN_ENABLE</code>	1	EXTTEST_TRAIN instruction enabled
<code>JTAG_EXTTEST_PULSE_ENABLE</code>	1	EXTTEST_PULSE instruction enabled

Parameter	Value	Description
JTAG_INTEST_ENABLE	1	INTEST instruction enabled
JTAG_CLAMP_ENABLE	1	CLAMP instruction enabled
JTAG_HIGHZ_ENABLE	1	HIGHZ instruction enabled
JTAG_RUNBIST_ENABLE	1	RUNBIST instruction enabled
JTAG_TMP_ENABLE	1	TMP controller enabled
JTAG_IC_RESET_ENABLE	1	Enables the external IC_RESET TDR slice (SMU drives this into the DTP's JTAG_IC_RESET_EXT_ENABLE). SMC and SEP slices are enabled unconditionally by the SMU.
JTAG_SMC_DBG_ENABLE	1	SMC debug interface enabled
JTAG_STAP_IO_ENABLE	1	I/O STAP enabled
JTAG_NUM_EXTRA_STAPS	1	Number of additional STAPs
JTAG_IDCODE_MFR_ID	110h000	Manufacturer ID
JTAG_IDCODE_PART_NUM	160h0000	Part number
JTAG_IDCODE_SI_REV	40h0	Silicon revision
JTAG_OCH_VER	80h00	OCH IP version
XTRIG_INT_CT_MODE	80h00	Internal CT protocol mode bits (0=pulse sync, 1=req/ack)
SMC_OTP_RD_PL_DEPTH	20h3	SMC OTP read pipeline depth
SMC_OTP_WR_PL_DEPTH	20h3	SMC OTP write pipeline depth
SMC_RD_PL_DEPTH	20h3	SMC fabric read pipeline depth
SMC_WR_PL_DEPTH	20h3	SMC fabric write pipeline depth

SEP-related DTP configuration is not exposed as an SMU config-struct field. The SMU derives SEP debug/STAP enablement from the module parameter `SEP` and wires the corresponding DTP interfaces internally to the SEP block when `SEP=1`.

The DTP IC_RESET TDR is composed of three independent struct-typed slices. The SMU passes `smc_pkg::jtag_smc_reset_ctrl_t` into the DTP's `ic_reset_smc_t` parameter (with `JTAG_IC_RESET_SMC_ENABLE=1`), `sep_pkg::jtag_sep_reset_ctrl_t` into `ic_reset_sep_t` (with `JTAG_IC_RESET_SEP_ENABLE=SEP`), and exposes the external slice via the SMU's own `ic_reset_ext_t` type parameter and `jtag_ic_reset_ext_o` port (gated by `JTAG_IC_RESET_ENABLE`). Each slice's `.ovrd/.val` sub-structs must be the same width; `jtag_ptap` interleaves them into the `{enable, control}` bit order required by the JTAG specification.

1.4.5. DTP Parameter Pass-Through

The SMU passes through most DTP parameters to allow integrator customization. Type parameters are set internally based on SMC package types.

JTAG Configuration Parameters

Parameter	SMU Default	Pass-through	Notes
JTAG_BSR_ENABLE	1	Yes	Boundary scan enabled
JTAG_EXTEST_TRAIN_ENABLE	1	Yes	
JTAG_EXTEST_PULSE_ENABLE	1	Yes	
JTAG_INTEST_ENABLE	1	Yes	
JTAG_CLAMP_ENABLE	1	Yes	

Parameter	SMU Default	Pass-through	Notes
JTAG_HIGHZ_ENABLE	1	Yes	
JTAG_RUNBIST_ENABLE	1	Yes	
JTAG_TMP_ENABLE	1	Yes	
JTAG_IC_RESET_ENABLE	1	No	SMU routes this to the DTP's JTAG_IC_RESET_EXT_ENABLE; SMC and SEP slices are enabled inside the SMU instantiation (SMC unconditionally, SEP when SEP=1)
JTAG_SMC_DBG_ENABLE	1	Yes	
JTAG_SEP_DBG_ENABLE	1 (SEP=1)	No	Derived internally from the SMU module parameter SEP; not exposed in smu_cfg_t
JTAG_STAP_IO_ENABLE	1	Yes	
JTAG_NUM_EXTRA_STAPS	1	Yes	Port array size equals parameter value; when set to 0, array size is 1 (tie-off required)

JTAG ID Parameters

Parameter	SMU Default	Pass-through	Notes
JTAG_IDCODE_MFR_ID	110h000	Yes	Manufacturer ID
JTAG_IDCODE_PART_NUM	160h0000	Yes	Part number
JTAG_IDCODE_SI_REV	40h0	Yes	Silicon revision
JTAG_OCH_VER	80h00	Yes	OCH IP version

Cross Trigger Parameters

Parameter	SMU Default	DTP Internal	Pass-through	Notes
XTRIG_INT_CT_MODE	[7:0]	[9:0]	Yes	8-bit on SMU; DTP bits [1:0]=0 for SMC

The SMU parameter width is two less than DTP's for cross trigger (8 vs 10) because internal cross trigger ports [1:0] and clock stop request #0 are connected to the SMC internally. The SMU remaps DTP ports [9:2] to SMU external ports [7:0].

Pipeline Depth Parameters

Parameter	SMU Default	Pass-through	Notes
SMC_OTP_RD_PL_DEPTH	20h3	Yes	SMC OTP read pipeline depth
SMC_OTP_WR_PL_DEPTH	20h3	Yes	SMC OTP write pipeline depth
SEP_OTP_RD_PL_DEPTH	20h3	No	Fixed to 20h3 in SMU; used when SEP=1, ignored when SEP=0
SEP_OTP_WR_PL_DEPTH	20h3	No	Fixed to 20h3 in SMU; used when SEP=1, ignored when SEP=0
SMC_RD_PL_DEPTH	20h3	Yes	SMC fabric read pipeline depth
SMC_WR_PL_DEPTH	20h3	Yes	SMC fabric write pipeline depth

Type Parameters

The following type parameters are set internally by the SMU and are not exposed for integrator configuration.

Parameter	Set To	Notes
jtag_tap_ctrl_t	prim_jtag_pkg::jtag_tap_ctrl_t	JTAG TAP control struct
jtag_scan_ctrl_t	prim_jtag_pkg::jtag_scan_ctrl_t	JTAG scan control struct

Parameter	Set To	Notes
smc_jtag_axi_req_t	smc_pkg::smc_jtag_56_64_2_12_axi_req_t	SMC debug AXI request
smc_jtag_axi_resp_t	smc_pkg::smc_jtag_56_64_2_12_axi_resp_t	SMC debug AXI response
smc_otp_axil_req_t	smc_efuse_pkg::efuse_axil_req_t	SMC OTP AXI-Lite request
smc_otp_axil_resp_t	smc_efuse_pkg::efuse_axil_resp_t	SMC OTP AXI-Lite response
sep_otp_axil_req_t	smc_efuse_pkg::efuse_axil_req_t	SEP OTP AXI-Lite request (error slave when SEP=0)
sep_otp_axil_resp_t	smc_efuse_pkg::efuse_axil_resp_t	SEP OTP AXI-Lite response (error slave when SEP=0)
xtrig_axil_req_t	smc_pkg::smc_axil_32_32_req_t	Cross trigger CSR request
xtrig_axil_resp_t	smc_pkg::smc_axil_32_32_resp_t	Cross trigger CSR response

1.4.6. DTP Port Mapping

The following tables define the complete DTP port list as connected within the SMU. Ports marked "SMU external port" are exposed on the SMU boundary for integrator connection. Ports marked with a subsystem name (e.g. SMC, SEP) are wired internally within the SMU.

Clock and Reset

Port	Direction	Width	SMU Connection
clk_i	input	1	SMU clk_smu_i (shared with SMC clk_smc_i)
rst_n_i	input	1	SMC rst_primary_smc_clk_no
pwr_on_rst_ni	input	1	SMU rst_cold_ni (power-on reset for JTAG logic)

Primary JTAG TAP Interface

Port	Direction	Width	SMU Connection
jtag_ptap_client_tap_ctrl_i	input	jtag_tap_ctrl_t	SMU external port
jtag_ptap_client_tdi_i	input	1	SMU external port
jtag_ptap_client_tdo_o	output	1	SMU external port
jtag_ptap_client_tdo_oen_o	output	1	SMU external port

Boundary Scan Interface

Port	Direction	Width	SMU Connection
jtag_bsr_host_scan_ctrl_o	output	jtag_scan_ctrl_t	SMU external port
jtag_bsr_host_scan_in_i	input	1	SMU external port
jtag_bsr_host_scan_out_o	output	1	SMU external port

I/O STAP Interface

Port	Direction	Width	SMU Connection
jtag_stap_io_host_tap_ctrl_o	output	jtag_tap_ctrl_t	SMU external port
jtag_stap_io_host_tdi_i	input	1	SMU external port
jtag_stap_io_host_tdo_o	output	1	SMU external port
jtag_stap_io_host_tdo_oen_o	output	1	SMU external port

SMC Debug STAP Interface

Port	Direction	Width	SMU Connection
jtag_stap_smc_host_tap_ctrl_o	output	jtag_tap_ctrl_t	SMC smc_cpu_jtag_{TCK,TMS}_i; trst_n inverted drives SMC smc_cpu_jtag_reset_i when JTAG_SMC_DBG_ENABLE=1, else tie-off
jtag_stap_smc_host_tdi_i	input	1	SMC smc_cpu_jtag_TDO_data_o when JTAG_SMC_DBG_ENABLE=1, else 1 b0
jtag_stap_smc_host_tdo_o	output	1	SMC smc_cpu_jtag_TDI_i when JTAG_SMC_DBG_ENABLE=1, else unused
jtag_stap_smc_host_tdo_oen_o	output	1	Internal only; not consumed by SMC

SEP Debug STAP Interface

Port	Direction	Width	SMU Connection
jtag_stap_sep_host_tap_ctrl_o	output	jtag_tap_ctrl_t	SEP jtag_{tck,tms,trst_n} when SEP=1, else unused
jtag_stap_sep_host_tdi_i	input	1	SEP jtag_tdo when SEP=1, else 1 b0
jtag_stap_sep_host_tdo_o	output	1	SEP jtag_tdi when SEP=1, else unused
jtag_stap_sep_host_tdo_oen_o	output	1	Internal only; not consumed by SEP at SMU level

Extra STAP Interfaces

Port	Direction	Width	SMU Connection
jtag_stap_extra_host_tap_ctrl_o	output	jtag_tap_ctrl_t[1:0]	SMU external port
jtag_stap_extra_host_tdi_i	input	[1:0]	SMU external port
jtag_stap_extra_host_tdo_o	output	[1:0]	SMU external port
jtag_stap_extra_host_tdo_oen_o	output	[1:0]	SMU external port

Extended STAP Scan Interface

Port	Direction	Width	SMU Connection
jtag_stap_host_scan_ctrl_o	output	jtag_scan_ctrl_t	SMU external port
jtag_stap_host_scan_in_i	input	1	SMU external port
jtag_stap_host_scan_out_o	output	1	SMU external port

DFD iJTAG Interface

Port	Direction	Width	SMU Connection
jtag_dfd_host_scan_ctrl_o	output	jtag_scan_ctrl_t	SMU external port
jtag_dfd_host_scan_in_i	input	1	SMU external port
jtag_dfd_host_scan_out_o	output	1	SMU external port

DFT iJTAG Interface

Port	Direction	Width	SMU Connection
jtag_dft_host_scan_ctrl_o	output	jtag_scan_ctrl_t	SMU external port
jtag_dft_host_scan_in_i	input	1	SMU external port
jtag_dft_host_scan_out_o	output	1	SMU external port

SMC Fabric Debug AXI Interface

Port	Direction	Width	SMU Connection
axi_smc_dbg_req_o	output	smc_jtag_axi_req_t	SMC jtag_axi_in_req_i
axi_smc_dbg_resp_i	input	smc_jtag_axi_resp_t	SMC jtag_axi_in_resp_o

SMC OTP Debug AXI-Lite Interface

Port	Direction	Width	SMU Connection
axil_smc_otp_jtag_req_o	output	smc_otp_axil_req_t	SMC axil_smc_otp_jtag_req_i
axil_smc_otp_jtag_resp_i	input	smc_otp_axil_resp_t	SMC axil_smc_otp_jtag_resp_o

SEP OTP Debug AXI-Lite Interface

When `SEP=1`, this interface connects to the SEP `axil_sep_otp_jtag_*` ports. When `SEP=0`, the SMU instantiates `u_sep_otp_axil_err_slv`, which responds with DECERR to all transactions.

Port	Direction	Width	SMU Connection
axil_sep_otp_jtag_req_o	output	sep_otp_axil_req_t	SEP <code>axil_sep_otp_jtag_req_i</code> when <code>SEP=1</code> , else AXI-Lite error slave (<code>u_sep_otp_axil_err_slv</code>)
axil_sep_otp_jtag_resp_i	input	sep_otp_axil_resp_t	SEP <code>axil_sep_otp_jtag_resp_o</code> when <code>SEP=1</code> , else AXI-Lite error slave (<code>u_sep_otp_axil_err_slv</code>)

Clock Control

Port	Direction	Width	SMU Connection
stop_clks_o	output	1	SMU external port: <code>dtb_stop_clks_o</code> (to PLL clock gates)

Boot Stall Control

Port	Direction	Width	SMU Connection
jtag_boot_stall_ovrd_o	output	1	Disconnected
jtag_boot_stall_o	output	1	Disconnected

JTAG Reset Control

Port	Direction	Width	SMU Connection
jtag_ic_reset_smc_o	output	smc_pkg::jtag_smc_reset_ctrl_t	SMC <code>jtag_smc_reset_ctrl_i</code> (packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values)
jtag_ic_reset_sep_o	output	sep_pkg::jtag_sep_reset_ctrl_t	SEP <code>jtag_sep_reset_ctrl_i</code> (packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values)
jtag_ic_reset_ext_o	output	ic_reset_ext_t	SMU <code>jtag_ic_reset_ext_o</code> external port (integrator-defined packed struct)

JTAG State Outputs

Port	Direction	Width	SMU Connection
jtag_ptap_state_o	output	tap_state_e	SMU external port

Port	Direction	Width	SMU Connection
jtag_ptap_inst_decoded_o	output	jtag_instruction_decoded_e	SMU external port

Cross Trigger CSR AXI-Lite Interface

Port	Direction	Width	SMU Connection
axil_xtrig_req_i	input	xtrig_axil_req_t	SMC axil_dtp_csr_req_o
axil_xtrig_resp_o	output	xtrig_axil_resp_t	SMC axil_dtp_csr_resp_i

Cross Trigger Matrix Interface

Port	Direction	Width	SMU Connection
xtrig_ctm_src_req_o	output	[XTRIG_NUM_INT_CT-1:0]	[1:0]: SMU internal (SMC), [9:2]: SMU external xtrig_ctm_src_req_o[7:0]
xtrig_ctm_src_ack_i	input	[XTRIG_NUM_INT_CT-1:0]	[1:0]: Tie-off (SMC), [9:2]: SMU external xtrig_ctm_src_ack_i[7:0]
xtrig_ctm_dst_req_i	input	[XTRIG_NUM_INT_CT-1:0]	[1:0]: SMC xtrigger_ss_o[1:0], [9:2]: SMU external xtrig_ctm_dst_req_i[7:0]
xtrig_ctm_dst_ack_o	output	[XTRIG_NUM_INT_CT-1:0]	[1:0]: Unused (SMC connection), [9:2]: SMU external xtrig_ctm_dst_ack_o[7:0]

Internal cross trigger ports [1:0] are reserved for SMC (both bits of `xtrigger_ss_o`). The SMU exposes ports [9:2] as external ports [7:0], so the SMU `XTRIG_NUM_INT_CT` parameter is two less than DTP's (8 vs 10). The SMC uses pulse-synchronization mode, so ack signals for ports [1:0] are unused: `xtrig_ctm_src_ack_i[1:0]` should be tied to `2'b00` and `xtrig_ctm_dst_ack_o[1:0]` are left unconnected. The usage of ack signals depends on the `XTRIG_INT_CT_MODE` parameter: when a mode bit is 0 (pulse synchronization), the corresponding ack is unused; when 1 (req/ack handshaking), the ack must be connected to the cross trigger endpoint.

Clock Stop Request Interface

Port	Direction	Width	SMU Connection
xtrig_clk_stop_req_i	input	[XTRIG_NUM_CLK_STOP_REQ-1:0]	[0]: SMC tdr_dbg_ctrl_clocks_stopped_by_cla_o, [8:1]: SMU external xtrig_clk_stop_req_i[7:0]

Clock stop request port #0 is reserved for SMC. The SMU exposes ports [8:1] as external ports [7:0], so the SMU `XTRIG_NUM_CLK_STOP_REQ` parameter should be one less than DTP's (8 vs 9).

Cross Trigger Port GPIO Interface

The 16 CTP signals from DTP are ported through the SMU to `smu_wrapper` as external ports. In `smu_wrapper`, they connect to internal signals destined for `smc_ip_integration`. Currently, input signals are tied off to '0 and output signals are dangling pending proper CTP-to-GPIO connection implementation.

Port	Direction	Width	SMU Connection
xtrig_ctp_req_out_dout_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_req_out_dout_en_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_req_out_din_i	input	[15:0]	SMU external → smu_wrapper internal (tied to '0)
xtrig_ctp_req_out_din_en_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_req_in_dout_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)

Port	Direction	Width	SMU Connection
xtrig_ctp_req_in_dout_en_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_req_in_din_i	input	[15:0]	SMU external → smu_wrapper internal (tied to '0')
xtrig_ctp_req_in_din_en_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_ack_in_dout_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_ack_in_dout_en_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_ack_in_din_i	input	[15:0]	SMU external → smu_wrapper internal (tied to '0')
xtrig_ctp_ack_in_din_en_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_ack_out_dout_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_ack_out_dout_en_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)
xtrig_ctp_ack_out_din_i	input	[15:0]	SMU external → smu_wrapper internal (tied to '0')
xtrig_ctp_ack_out_din_en_o	output	[15:0]	SMU external → smu_wrapper internal (dangling)

1.4.7. SMC Port Mapping

The following tables define the complete SMC port list as connected within the SMU. Ports marked "SMU external port" are exposed on the SMU boundary. Ports marked with "DTP" or "SEP" are wired internally.

Clocks

Port	Direction	Width	SMU Connection
clk_smc_i	input	1	SMU <code>clk_smu_i</code> (shared with DTP <code>clk_i</code>)
clk_periph_i	input	1	SMU external port
clk_ref_i	input	1	SMU external port

Resets

Port	Direction	Width	SMU Connection
rst_cold_ni	input	1	SMU external port
rst_primary_ref_clk_no	output	1	SMU external port
rst_primary_smc_clk_no	output	1	SMU external port + DTP <code>rst_n_i</code>

System AXI Input

Port	Direction	Width	SMU Connection
sys_axi_in_req_i	input	<code>smc_sys_in_56_64_2_12_axi_req_t</code>	SMU external port
sys_axi_in_resp_o	output	<code>smc_sys_in_56_64_2_12_axi_resp_t</code>	SMU external port

JTAG AXI Input

Port	Direction	Width	SMU Connection
jtag_axi_in_req_i	input	<code>smc_jtag_56_64_2_12_axi_req_t</code>	DTP <code>axi_smc_dbg_req_o</code>
jtag_axi_in_resp_o	output	<code>smc_jtag_56_64_2_12_axi_resp_t</code>	DTP <code>axi_smc_dbg_resp_i</code>

SMC OTP JTAG AXI-Lite

Port	Direction	Width	SMU Connection
axil_smc_otp_jtag_req_i	input	efuse_axil_req_t	DTP axil_smc_otp_jtag_req_o
axil_smc_otp_jtag_resp_o	output	efuse_axil_resp_t	DTP axil_smc_otp_jtag_resp_i

SEP AXI Input

When **SEP=0**, the SEP AXI subordinate input is tied off to '0' and the response output is left unconnected. When **SEP=1**, the SMU drives this path from the SEP-to-SMC dedicated AXI connection.

Port	Direction	Width	SMU Connection
sep_axi_in_req_i	input	smc_sep_in_56_64_6_12_axi_req_t	Tied off to '0' (no incoming transactions)
sep_axi_in_resp_o	output	smc_sep_in_56_64_6_12_axi_resp_t	Unconnected (no transactions to respond to)

AXI Output

Port	Direction	Width	SMU Connection
output_axi_req_o	output	smc_sys_out_56_64_8_12_axi_req_t	SMU external port
output_axi_resp_i	input	smc_sys_out_56_64_8_12_axi_resp_t	SMU external port

DTP CSR Interface

Port	Direction	Width	SMU Connection
axil_dtp_csr_req_o	output	smc_axil_32_32_req_t	DTP axil_xtrig_req_i
axil_dtp_csr_resp_i	input	smc_axil_32_32_resp_t	DTP axil_xtrig_resp_o

PLL Interface AXI-Lite

Port	Direction	Width	SMU Connection
axil_pll_req_o	output	smc_axil_32_32_req_t	SMU external port (to PLL shim)
axil_pll_resp_i	input	smc_axil_32_32_resp_t	SMU external port (from PLL shim)

PVT Interface AXI-Lite

Port	Direction	Width	SMU Connection
axil_pvt_req_o	output	smc_axil_32_32_req_t	SMU external port (to PVT shim)
axil_pvt_resp_i	input	smc_axil_32_32_resp_t	SMU external port (from PVT shim)

eFuse Interface

Port	Direction	Width	SMU Connection
efuse_bank_ctrl_req_o	output	efuse_axil_req_t	SMU external port
efuse_bank_ctrl_resp_i	input	efuse_axil_resp_t	SMU external port
efuse_shim_command_req_o	output	efuse_command_req_t	SMU external port
efuse_shim_command_resp_i	input	efuse_command_resp_t	SMU external port

Port	Direction	Width	SMU Connection
shadow_regs_o	output	efuse_map_t	SMU external port

GPIO Shim Interface

Port	Direction	Width	SMU Connection
axil_req_gpio_ctrl_o	output	gpio_axil_req_t	SMU external port
axil_resp_gpio_ctrl_i	input	gpio_axil_resp_t	SMU external port

GPIO Data Signals

Port	Direction	Width	SMU Connection
lsio_interface_select_o	output	[NUM_GPIO_WRAPS-1:0]	SMU external port
gpio_in_i	input	[NUM_GPIO_WRAPS-1:0]	SMU external port
gpio_out_o	output	[NUM_GPIO_WRAPS-1:0]	SMU external port
gpio_in_en_o	output	[NUM_GPIO_WRAPS-1:0]	SMU external port
gpio_out_en_o	output	[NUM_GPIO_WRAPS-1:0]	SMU external port

GPIO External Pins

Port	Direction	Width	SMU Connection
rst_cool_n_from_pin_i	input	1	SMU external port

ATB Telemetry

Port	Direction	Width	SMU Connection
clk_telemetry_i	input	1	SMU external port
rst_telemetry_ni	input	1	SMU external port
telemetry_atdata_i	input	telemetry_data_t[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_atid_i	input	atb_id_t[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_atready_o	output	[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_atvalid_i	input	[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_afvalid_o	output	[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_afready_i	input	[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port

DED/WDT

Port	Direction	Width	SMU Connection
cluster_ded_o	output	1	SMU external port
wdt_first_timeout_o	output	1	SMU external port
wdt_second_timeout_o	output	1	SMU external port

Region Size

Port	Direction	Width	SMU Connection
smc_region_size_o	output	[31:0]	SMU external port

External Interrupts

Port	Direction	Width	SMU Connection
ext_interrupts_i	input	[NUM_EXT_INTERRUPTS-1:0]	SMU external port
sep_mailbox_interrupts_i	input	[7:0]	Internal: from SEP smc_mailbox_interrupt_o when SEP=1, else tied to '0
sep_wdt_reset_n_i	input	1	Internal: ~sep_wdt_timer_rst_req when SEP=1, else tied to 1b1

Fuse Signals

Port	Direction	Width	SMU Connection
fuse_sense_done_o	output	1	SMU external port
fuse_reset_n_delayed_o	output	1	SMU external port

DFT Signals

Port	Direction	Width	SMU Connection
skip_mem_repair_o	output	1	SMU external port
ext_boot_seq_done_i	input	1	SMU external port

PVT

Port	Direction	Width	SMU Connection
temp_interrupt_i	input	1	SMU external port

Lifecycle State

Port	Direction	Width	SMU Connection
lc_state_o	output	[2*LC_STATE_WIDTH-1:0]	SEP lifecycle state when SEP=1, else 80hf0; ported through SMU
lc_sigint_err_o	output	1	SEP lifecycle integrity error when SEP=1, else SMC efuse_lc_sigint_err

RAS Bank Settings

Port	Direction	Width	SMU Connection
ras_bank_chip_o	output	[3:0]	SMU external port
ras_bank_instance_o	output	[3:0]	SMU external port

Mailbox Interrupts

Port	Direction	Width	SMU Connection
ext_mailbox_interrupts_o	output	[NUM_MAILBOXES-1:0]	SMU external port

Reset Unit Signals

Port	Direction	Width	SMU Connection
cfg_flr_pf_active_i	input	1	SMU external port
isolate_req_o	output	[31:0]	SMU external port
ss_reset_complete_i	input	[31:0]	SMU external port
ss_config_o	output	[31:0]	SMU external port
ss_reset_ctrl_o	output	reset_ctrl_t[31:0]	SMU external port
sync_irq_o	output	1	SMU external port

CPU Memory Signals

Port	Direction	Width	SMU Connection
rom_intf_req_o	output	rom_req_t	SMU external port
rom_intf_rsp_i	input	rom_rsp_t	SMU external port
scratch_ram_intf_req_o	output	scratch_ram_req_t[31:0]	SMU external port
scratch_ram_intf_rsp_i	input	scratch_ram_rsp_t[31:0]	SMU external port
l1_icache_tag_intf_req_o	output	l1_icache_tag_req_t[3:0]	SMU external port
l1_icache_tag_intf_rsp_i	input	l1_icache_tag_rsp_t[3:0]	SMU external port
l1_icache_data_intf_req_o	output	l1_icache_data_req_t[7:0]	SMU external port
l1_icache_data_intf_rsp_i	input	l1_icache_data_rsp_t[7:0]	SMU external port
l1_dcache_tag_intf_req_o	output	l1_dcache_tag_req_t[3:0]	SMU external port
l1_dcache_tag_intf_rsp_i	input	l1_dcache_tag_rsp_t[3:0]	SMU external port
l1_dcache_data_intf_req_o	output	l1_dcache_data_req_t[3:0]	SMU external port
l1_dcache_data_intf_rsp_i	input	l1_dcache_data_rsp_t[3:0]	SMU external port

Memory Init

Port	Direction	Width	SMU Connection
disable_sram_auto_init_i	input	1	SMU external port
init_mem_done_o	output	1	SMU external port

System Timer OCTS Interface

Port	Direction	Width	SMU Connection
timer_count_o	output	[63:0]	SMU external port

Reset Overrides (from DTP)

Port	Direction	Width	SMU Connection
jtag_reset_ctrl_i	input	smc_pkg::jtag_smc_reset_ctrl_t	DTP jtag_ic_reset_smc_o (packed struct with .ovrd override enables and .val active-low override values)

DFD Signals

Port	Direction	Width	SMU Connection
<code>cla_gpio_enable_o</code>	output	1	SMU external port
<code>tdr_dbg_ctrl_clock_stop_en_i</code>	input	1	DTP <code>cla_clock_stop_en_o</code>
<code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code>	output	1	DTP <code>xtrig_clk_stop_req_i[0]</code> (SMU connects SMC output to DTP input)
<code>halt_clock_global_i</code>	input	[NUM_MAX_EXTERNAL_XTRIG_AGENTS-1:0]	SMU external port

Cross Trigger Interface

Port	Direction	Width	SMU Connection
<code>xtrigger_pads_i</code>	input	<code>xtrigger_t</code>	SMU external port
<code>xtrigger_pads_o</code>	output	<code>xtrigger_t</code>	SMU external port
<code>xtrigger_ss_i</code>	input	<code>xtrigger_t</code> [NUM_MAX_EXTERNAL_XTRIG_AGENTS-1:0]	SMU external port
<code>xtrigger_ss_o</code>	output	<code>xtrigger_t</code>	[1:0]: DTP <code>xtrig_ctm_dst_req_i[1:0]</code> (both bits to DTP)

Trace Memory

Port	Direction	Width	SMU Connection
<code>trace_mem_req_o</code>	output	<code>SinkMemPktIn_s</code> [TRC_RAM_INSTANCES-1:0]	SMU external port
<code>trace_mem_resp_i</code>	input	<code>SinkMemPktOut_s</code> [TRC_RAM_INSTANCES-1:0]	SMU external port

External Debug Bus

Port	Direction	Width	SMU Connection
<code>ext_debug_bus_i</code>	input	[511:0]	SMU external port

Test Mode

Port	Direction	Width	SMU Connection
<code>test_en_i</code>	input	1	SMU external port
<code>scan_rst_ni</code>	input	1	SMU external port

Captured Straps

Port	Direction	Width	SMU Connection
<code>captured_straps_i</code>	input	[63:0]	SMU external port

Please refer to the `reset_unit` STRAPS_LO and STRAPS_HI register descriptions for the decoded value of the straps.

1.4.8. SMC-DTP Internal Interconnection

The following table summarizes the key internal connections between the DTP and SMC within the SMU. These connections are not visible on the SMU boundary.

Connection	DTP Port	SMC Port	Direction	Notes
AXI Debug Bus	<code>axi_smc_dbg_req_o</code>	<code>jtag_axi_in_req_i</code>	DTP → SMC	
	<code>axi_smc_dbg_resp_i</code>	<code>jtag_axi_in_resp_o</code>	SMC → DTP	
AXI OTP Bus	<code>axil_smc_otp_jtag_req_o</code>	<code>axil_smc_otp_jtag_req_i</code>	DTP → SMC	
	<code>axil_smc_otp_jtag_resp_i</code>	<code>axil_smc_otp_jtag_resp_o</code>	SMC → DTP	
DTP CSR Bus	<code>axil_xtrig_req_i</code>	<code>axil_dtp_csr_req_o</code>	SMC → DTP	
	<code>axil_xtrig_resp_o</code>	<code>axil_dtp_csr_resp_i</code>	DTP → SMC	
JTAG SMC Reset Overrides	<code>jtag_ic_reset_smc_o</code>	<code>jtag_reset_ctrl_i</code>	DTP → SMC	Packed <code>smc_pkg::jtag_smc_reset_ctrl_t</code> struct carrying all SMC reset override enables and active-low values
DTP Reset	<code>rst_n_i</code>	<code>rst_primary_smc_clk_no</code>	SMC → DTP	
Cross Trigger [1:0]	<code>xtrig_ctm_dst_req_i[1:0]</code>	<code>xtrigger_ss_o[1:0]</code>	SMC → DTP	Both bits
Clock Stop #0	<code>xtrig_clk_stop_req_i[0]</code>	<code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code>	SMC → DTP	
Boot Stall	<code>jtag_boot_stall_ovrd_o</code>	Disconnected	N/A	
	<code>jtag_boot_stall_o</code>	Disconnected	N/A	

1.4.9. SEP-Conditional Wiring

SEP-related interfaces are handled internally within the SMU. Their behavior depends on the module parameter `SEP`.

Module/Signal	Implementation	Description
SMC SEP AXI Input	<code>sep_axi_in_req_i</code> = SEP traffic when <code>SEP=1</code> , else '0	Dedicated SEP-to-SMC subordinate path
DTP SEP OTP JTAG	SEP <code>axil_sep_otp_jtag_*</code> when <code>SEP=1</code> , else <code>u_sep_otp_axil_err_slv</code>	SEP OTP debug AXI-Lite path
DTP SEP STAP return	SEP <code>jtag_tdo</code> when <code>SEP=1</code> , else <code>sep_stap_tdo_to_dtp = 1'b0</code>	SEP JTAG STAP return path
SMU	<code>sep_mailbox_interrupts</code>	Internal SEP-to-SMC mailbox interrupts when <code>SEP=1</code> , else '0
SMC	<code>sep_wdt_reset_n_i</code> = <code>~sep_wdt_timer_rst_req</code> when <code>SEP=1</code> , else 1'b1	SEP watchdog reset input to SMC

Other Internal Tie-offs

Module	Port	Tie-off Value	Reason
DTP	<code>xtrig_ctm_src_ack_i[1:0]</code>	2'b00	SMC uses pulse-sync mode

`xtrig_ctm_src_ack_i[9:2]` (exposed as SMU `xtrig_ctm_src_ack_i[7:0]`) is ported out through the SMU. The integrator should tie off to '0 when using pulse-sync mode (`XTRIG_INT_CT_MODE` bit = 0), or connect to actual ack signals when using req/ack mode (bit = 1).

1.4.10. SMU Port Interface

The SMU exposes a single consolidated boundary that an adopter connects to at the chip or chiplet level. The following table defines all SMU top-level ports including clock, reset, JTAG, cross trigger, SMC-originated interfaces (AXI, GPIO, eFuse, PLL, PVT, memory), and SEP passthrough interfaces (memory macros, eFuse, crypto SRAM, SPI, AXI extension). Ports prefixed with `sep_` are only active when the module parameter `SEP=1`.

Table 7. SMU Port Declaration

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
clk_smu_i	logic	Input	1	Primary SMU clock domain. Shared by SMC and DTP system logic.	PLL output	N/A
clk_ref_i	logic	Input	1	Reference clock domain for timing-sensitive operations.	PLL reference clock	N/A
rst_cold_ni	logic	Input	1	Cold reset (active low). Asynchronous assertion, synchronous deassertion. Also serves as DTP power-on reset.	System cold reset	N/A
rst_cold_stable_ref_clk_no	logic	Output	1	Stable cold reset synchronized to reference clock domain (active low).	Downstream subsystems	N/A
powergood_i	logic	Input	1	Power-good indication from supply supervisor.	Power supply supervisor	N/A
jtag_ptap_client_tap_ctrl_i	prim_jtag_pkg::jtag_tap_ctrl_t	Input	-	Primary JTAG TAP control (TCK, TMS, TRST_N).	External JTAG pads	N/A
jtag_ptap_client_tdi_i	logic	Input	1	Primary JTAG TDI.	External JTAG pads	N/A
jtag_ptap_client_tdo_o	logic	Output	1	Primary JTAG TDO.	External JTAG pads	N/A
jtag_ptap_client_tdo_oen_o	logic	Output	1	Primary JTAG TDO output enable.	External JTAG pads	N/A
jtag_bsr_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	Boundary scan control.	Boundary scan chain	N/A
jtag_bsr_host_scan_in_i	logic	Input	1	Boundary scan data in.	Boundary scan chain	N/A
jtag_bsr_host_scan_out_o	logic	Output	1	Boundary scan data out.	Boundary scan chain	N/A
jtag_stap_io_host_tap_ctrl_o	prim_jtag_pkg::jtag_tap_ctrl_t	Output	-	I/O STAP TAP control.	I/O STAP host	N/A
jtag_stap_io_host_tdi_i	logic	Input	1	I/O STAP TDI.	I/O STAP host	Tie to '0 if unused
jtag_stap_io_host_tdo_o	logic	Output	1	I/O STAP TDO.	I/O STAP host	N/A
jtag_stap_io_host_tdo_oen_o	logic	Output	1	I/O STAP TDO output enable.	I/O STAP host	N/A
jtag_stap_extra_host_tap_ctrl_o	prim_jtag_pkg::jtag_tap_ctrl_t	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TAP control array.	Extra STAP hosts	Tie to '0 if unused
jtag_stap_extra_host_tdi_i	logic	Input	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDI array.	Extra STAP hosts	Tie to '0 if unused
jtag_stap_extra_host_tdo_o	logic	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDO array.	Extra STAP hosts	N/A
jtag_stap_extra_host_tdo_oen_o	logic	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDO output enable array.	Extra STAP hosts	N/A
jtag_stap_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	Extended STAP scan control.	STAP scan chain	N/A
jtag_stap_host_scan_in_i	logic	Input	1	Extended STAP scan data in.	STAP scan chain	N/A
jtag_stap_host_scan_out_o	logic	Output	1	Extended STAP scan data out.	STAP scan chain	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
jtag_dfd_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	DFD iJTAG scan control.	DFD iJTAG chain	N/A
jtag_dfd_host_scan_in_i	logic	Input	1	DFD iJTAG scan data in.	DFD iJTAG chain	N/A
jtag_dfd_host_scan_out_o	logic	Output	1	DFD iJTAG scan data out.	DFD iJTAG chain	N/A
jtag_dft_secure_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	Secure DFT iJTAG scan control.	Secure DFT iJTAG chain	N/A
jtag_dft_secure_host_scan_in_i	logic	Input	1	Secure DFT iJTAG scan data in.	Secure DFT iJTAG chain	N/A
jtag_dft_secure_host_scan_out_o	logic	Output	1	Secure DFT iJTAG scan data out.	Secure DFT iJTAG chain	N/A
jtag_dft_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	Non-secure DFT iJTAG scan control.	Non-secure DFT iJTAG chain	N/A
jtag_dft_host_scan_in_i	logic	Input	1	Non-secure DFT iJTAG scan data in.	Non-secure DFT iJTAG chain	N/A
jtag_dft_host_scan_out_o	logic	Output	1	Non-secure DFT iJTAG scan data out.	Non-secure DFT iJTAG chain	N/A
dtp_stop_clks_o	logic	Output	1	Clock stop output from DTP (JTAG DEBUG_CONTROL or CLA clock-stop).	PLL clock gates	N/A
jtag_ptap_state_o	jtag_tap_pkg::tap_state_e	Output	-	Current PTAP TAP state.	Debug infrastructure	N/A
jtag_ptap_inst_decoded_o	jtag_inst_reg_pkg::jtag_instruction_decoded_e	Output	-	Current decoded PTAP instruction.	Debug infrastructure	N/A
jtag_ic_reset_ext_o	ic_reset_ext_t	Output	$\$bits(ic_reset_ext_t)$	External JTAG IC_RESET override slice. Integrator-defined packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values. Driven by the DTP's <code>jtag_ic_reset_ext_o</code> when <code>Cfg.JTAG_IC_RESET_ENABLE=1</code> , otherwise tied to '0.	Integrator-defined reset overrides	N/A
xtrig_ctm_src_req_o	logic	Output	$[XTRIG_NUM_INT_CT-1:0]$	Cross trigger source requests (SMU external ports [7:0]; DTP [1:0] reserved for SMC).	Cross trigger network	N/A
xtrig_ctm_src_ack_i	logic	Input	$[XTRIG_NUM_INT_CT-1:0]$	Cross trigger source acknowledges. Unused in pulse-sync mode (mode bit = 0).	Cross trigger network	Tie to '0 in pulse-sync mode
xtrig_ctm_dst_req_i	logic	Input	$[XTRIG_NUM_INT_CT-1:0]$	Cross trigger destination requests.	Cross trigger network	Tie to '0 if unused
xtrig_ctm_dst_ack_o	logic	Output	$[XTRIG_NUM_INT_CT-1:0]$	Cross trigger destination acknowledges. Unused in pulse-sync mode.	Cross trigger network	N/A
xtrig_clk_stop_req_i	logic	Input	$[XTRIG_NUM_CLK_STOP_REQ-1:0]$	Clock stop requests (SMU external ports [7:0]; port [0] reserved for SMC internally).	Clock stop requestors	Tie to '0 if unused
xtrig_ctp_req_out_dout_o	logic	Output	$[XTRIG_NUM_CTP-1:0]$	CTP request-out data output.	GPIO pad ring	N/A
xtrig_ctp_req_out_dout_en_o	logic	Output	$[XTRIG_NUM_CTP-1:0]$	CTP request-out data output enable.	GPIO pad ring	N/A
xtrig_ctp_req_out_din_i	logic	Input	$[XTRIG_NUM_CTP-1:0]$	CTP request-out data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_req_out_din_en_o	logic	Output	$[XTRIG_NUM_CTP-1:0]$	CTP request-out data input enable.	GPIO pad ring	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
xtrig_ctp_req_in_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data output.	GPIO pad ring	N/A
xtrig_ctp_req_in_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data output enable.	GPIO pad ring	N/A
xtrig_ctp_req_in_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP request-in data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_req_in_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data input enable.	GPIO pad ring	N/A
xtrig_ctp_ack_in_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data output.	GPIO pad ring	N/A
xtrig_ctp_ack_in_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data output enable.	GPIO pad ring	N/A
xtrig_ctp_ack_in_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP ack-in data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_ack_in_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data input enable.	GPIO pad ring	N/A
xtrig_ctp_ack_out_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data output.	GPIO pad ring	N/A
xtrig_ctp_ack_out_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data output enable.	GPIO pad ring	N/A
xtrig_ctp_ack_out_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP ack-out data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_ack_out_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data input enable.	GPIO pad ring	N/A
rst_primary_ref_clk_no	logic	Output	1	Primary reset synchronized to reference clock domain (active low).	Downstream subsystems	N/A
rst_primary_smc_clk_no	logic	Output	1	Primary reset synchronized to SMC clock domain (active low).	Downstream subsystems	N/A
smu_axi_in_req_i	smu_axi_xbar_pkg::axi_56_64_req_t	Input	-	SMU AXI crossbar inbound request (from SMN).	SMN fabric	Tie to '0 if unused
smu_axi_in_resp_o	smu_axi_xbar_pkg::axi_56_64_resp_t	Output	-	SMU AXI crossbar inbound response.	SMN fabric	N/A
smu_axi_out_req_o	smu_axi_xbar_pkg::axi_out_req_t	Output	-	SMU AXI crossbar outbound request (to SMN).	SMN fabric	N/A
smu_axi_out_resp_i	smu_axi_xbar_pkg::axi_out_resp_t	Input	-	SMU AXI crossbar outbound response.	SMN fabric	Tie to '0 if unused
axil_pll_req_o	smc_pkg::smc_axil_32_32_req_t	Output	-	PLL control AXI-Lite request.	PLL shim	N/A
axil_pll_resp_i	smc_pkg::smc_axil_32_32_resp_t	Input	-	PLL control AXI-Lite response.	PLL shim	N/A
axil_pvt_req_o	smc_pkg::smc_axil_32_32_req_t	Output	-	PVT monitoring AXI-Lite request.	PVT shim	N/A
axil_pvt_resp_i	smc_pkg::smc_axil_32_32_resp_t	Input	-	PVT monitoring AXI-Lite response.	PVT shim	N/A
smc_external_req_o	smc_pkg::smc_axil_32_32_req_t	Output	-	SMC AXI-Lite external port request for adopter peripheral expansion.	Adopter peripherals	N/A
smc_external_resp_i	smc_pkg::smc_axil_32_32_resp_t	Input	-	SMC AXI-Lite external port response.	Adopter peripherals	Tie to DECERR if unused
smc_efuse_bank_ctrl_req_o	smc_pkg::smc_axil_32_32_req_t	Output	-	SMC eFuse bank control AXI-Lite request.	eFuse shim	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
smc_efuse_bank_ctrl_resp_i	smc_pkg::smc_axil_32_32_resp_t	Input	-	SMC eFuse bank control AXI-Lite response.	eFuse shim	N/A
smc_efuse_shim_command_req_o	smc_efuse_pkg::fuse_command_req_t	Output	-	SMC eFuse command interface request.	eFuse shim	N/A
smc_efuse_shim_command_resp_i	smc_efuse_pkg::fuse_command_resp_t	Input	-	SMC eFuse command interface response.	eFuse shim	N/A
smc_shadow_regs_o	smc_efuse_pkg::efuse_map_t	Output	-	SMC eFuse shadow register output.	External systems	N/A
pll_clk_obs_i	logic	Input	1	PLL clock observation input.	PLL shim	N/A
pll_clk_obs_en_i	logic	Input	1	PLL clock observation enable.	Control logic	N/A
pvt_process_clk_obs_i	logic	Input	1	PVT process monitor clock observation input.	PVT shim	N/A
pvt_process_clk_obs_en_i	logic	Input	1	PVT process monitor clock observation enable.	Control logic	N/A
cat_therm_i	logic	Input	1	CAT thermal sensor input.	CAT thermal sensor	Tie to '0 if unused
axil_req_gpio_ctrl_o	gpio_pkg::gpio_axil_req_t	Output	-	GPIO controller AXI-Lite request.	GPIO padring	N/A
axil_resp_gpio_ctrl_i	gpio_pkg::gpio_axil_resp_t	Input	-	GPIO controller AXI-Lite response.	GPIO padring	N/A
lsio_interface_select_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	LSIO interface selection.	GPIO padring	N/A
gpio_in_i	logic	Input	[NUM_GPIO_WRAPS-1:0]	GPIO input data from external pins.	External GPIO pins	N/A
gpio_out_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	GPIO output data to external pins.	External GPIO pins	N/A
gpio_in_en_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	GPIO input enable control.	GPIO padring	N/A
gpio_out_en_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	GPIO output enable control.	GPIO padring	N/A
rst_cool_n_from_pin_i	logic	Input	1	Cool reset input from GPIO pin.	GPIO pin	N/A
clk_telemetry_i	logic	Input	1	Telemetry clock domain for ATB data collection.	Telemetry clock source	N/A
rst_telemetry_n_i	logic	Input	1	Telemetry reset (active low).	Telemetry reset source	N/A
telemetry_atdata_i	telemetry_receiver_pkg::telemetry_data_t	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry data input array.	ATB sources	N/A
telemetry_atid_i	telemetry_receiver_pkg::atb_id_t	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry ID input array.	ATB sources	N/A
telemetry_atready_o	logic	Output	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry ready output.	ATB sources	N/A
telemetry_atvalid_i	logic	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry valid input.	ATB sources	N/A
telemetry_atflushvalid_o	logic	Output	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB flush valid output.	ATB sources	N/A
telemetry_atflushready_i	logic	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB flush ready input.	ATB sources	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
cluster_ded_o	logic	Output	1	Cluster double error detection output.	External systems	N/A
wdt_first_timeout_o	logic	Output	1	First watchdog timeout signal.	Reset unit	N/A
wdt_second_timeout_o	logic	Output	1	Second watchdog timeout signal.	External systems	N/A
smc_region_size_o	logic	Output	[31:0]	SMC address region size output.	External systems	N/A
ext_interrupts_i	logic	Input	[Cfg.NUM_INT_O_SMC-1:0]	External interrupt inputs to SMC.	External interrupt sources	Tie to '0 if unused
fuse_sense_done_o	logic	Output	1	SMC fuse sense completion output.	Memory repair logic	N/A
fuse_reset_n_delayed_o	logic	Output	1	Delayed fuse reset output.	External systems	N/A
skip_mem_repair_o	logic	Output	1	Skip memory repair output.	Memory repair logic	N/A
ext_boot_seq_done_i	logic	Input	1	External boot sequence completion input (memory repair and shadow register override done); gates reset release.	External boot/repair controller	N/A
temp_interrupt_i	logic	Input	1	Temperature interrupt input.	PVT wrap	Tie to '0 if unused
lc_state_o	logic	Output	[2*LC_STATE_WIDTH-1:0]	Lifecycle state output (driven by SEP when SEP=1, else 8'hf0).	External systems	N/A
lc_sigint_err_o	logic	Output	1	Lifecycle signal integrity error.	External systems	N/A
ras_bank_chip_o	logic	Output	[3:0]	RAS bank chip selection output.	RAS system	N/A
ras_bank_instance_o	logic	Output	[3:0]	RAS bank instance selection output.	RAS system	N/A
ndmreset_request_i	logic	Input	[CPU_CLUSTER_COUNT-1:0]	Non-debug-module reset requests.	Debug infrastructure	Tie to '0 if unused
ndmreset_process_o	logic	Output	[CPU_CLUSTER_COUNT-1:0]	Non-debug-module reset process indicators.	Debug infrastructure	N/A
ext_mailbox_interrupts_o	logic	Output	[NUM_MAILBOXES-1:0]	External mailbox interrupt outputs.	External targets	N/A
cfg_flr_pf_active_i	logic	Input	1	Function Level Reset (FLR) PF active signal.	PCIe / system	Tie to '0 if unused
isolate_req_o	logic	Output	[31:0]	Isolation request output for 32 subsystems.	Subsystems	N/A
ss_reset_completion_i	logic	Input	[31:0]	Subsystem reset completion status.	Subsystems	32'hFFFFFFFF
ss_config_o	logic	Output	[31:0]	Subsystem configuration output.	Subsystems	N/A
ss_reset_ctrl_o	smc_reset_unit_pkg::reset_ctrl_t	Output	[31:0]	Subsystem reset control output array.	Subsystems	N/A
sync_irq_o	logic	Output	1	Synchronized interrupt output.	External systems	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
gpio_interrupt_o	logic	Output	[NUM_GPIO_WRAP-1:0]	Raw GPIO interrupt vector from the padding, one bit per GPIO wrap (bonded + unbonded). Passed through from SMC. Software reads GPIO status registers to identify the interrupt source.	SMC → external consumers	N/A, leave floating if unused
uart_interrupt_o	logic	Output	[NUM_UART-1:0]	Raw UART interrupt outputs, one bit per UART instance, in the peripheral clock domain. Passed through from SMC.	SMC → external consumers	N/A, leave floating if unused
rom_intf_req_o	rom_req_t	Output	-	SMC ROM memory interface request.	Boot ROM	N/A
rom_intf_rsp_i	rom_rsp_t	Input	-	SMC ROM memory interface response.	Boot ROM	N/A
scratch_ram_intf_req_o	scratch_ram_req_t	Output	[31:0]	SMC scratch RAM interface requests.	Scratch RAM banks	N/A
scratch_ram_intf_rsp_i	scratch_ram_rsp_t	Input	[31:0]	SMC scratch RAM interface responses.	Scratch RAM banks	N/A
l1_icache_tag_intf_req_o	l1_icache_tag_req_t	Output	[3:0]	SMC L1 I-cache tag interface requests.	I-cache tag banks	N/A
l1_icache_tag_intf_rsp_i	l1_icache_tag_rsp_t	Input	[3:0]	SMC L1 I-cache tag interface responses.	I-cache tag banks	N/A
l1_icache_data_intf_req_o	l1_icache_data_req_t	Output	[7:0]	SMC L1 I-cache data interface requests.	I-cache data banks	N/A
l1_icache_data_intf_rsp_i	l1_icache_data_rsp_t	Input	[7:0]	SMC L1 I-cache data interface responses.	I-cache data banks	N/A
l1_dcache_tag_intf_req_o	l1_dcache_tag_req_t	Output	[3:0]	SMC L1 D-cache tag interface requests.	D-cache tag banks	N/A
l1_dcache_tag_intf_rsp_i	l1_dcache_tag_rsp_t	Input	[3:0]	SMC L1 D-cache tag interface responses.	D-cache tag banks	N/A
l1_dcache_data_intf_req_o	l1_dcache_data_req_t	Output	[3:0]	SMC L1 D-cache data interface requests.	D-cache data banks	N/A
l1_dcache_data_intf_rsp_i	l1_dcache_data_rsp_t	Input	[3:0]	SMC L1 D-cache data interface responses.	D-cache data banks	N/A
disable_sram_auto_init_i	logic	Input	1	Disable SRAM automatic initialization.	Control logic	1'b0
init_mem_done_o	logic	Output	1	Memory initialization completion signal.	Control logic	N/A
chiplet_is_primary_i	logic	Input	1	Chiplet is primary (for System Timer OCTS).	GPIO padding straps	N/A
timer_count_o	logic	Output	[63:0]	64-bit system timer count output (OCTS).	External systems	N/A
trace_mem_req_o	dfd_trace_mem_pkg::SinkMemPktIn_s	Output	[TRC_RAM_INSTANCES-1:0]	Trace memory interface requests.	Trace memory	N/A
trace_mem_resp_i	dfd_trace_mem_pkg::SinkMemPktOut_s	Input	[TRC_RAM_INSTANCES-1:0]	Trace memory interface responses.	Trace memory	N/A
test_en_i	logic	Input	1	DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.	Test controller	1'b0
scan_rst_ni	logic	Input	1	Scan reset (active-low). Bypasses reset synchronizers during DFT.	Test controller	1'b1
captured_straps_i	logic	Input	[63:0]	Captured strap configuration. Please refer to the reset_unit STRAPS_LO and STRAPS_HI register descriptions for the decoded value of the straps.	GPIO padding	N/A
mem_repair_done_i	logic	Input	1	Memory repair done indicator.	DFT controller	1'b0

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
mem_repair_success_i	logic	Input	1	Memory repair success indicator.	DFT controller	1'b0
mem_repair_abort_i	logic	Input	1	Memory repair abort indicator.	DFT controller	1'b0
mbist_done_i	logic	Input	1	MBIST done indicator.	DFT controller	1'b0
mbist_pass_i	logic	Input	1	MBIST pass indicator.	DFT controller	1'b0
mbist_abort_i	logic	Input	1	MBIST abort indicator.	DFT controller	1'b0
sep_sram_req_o	sep_pkg::sep_sram_req_t	Output	-	SEP scratch SRAM request passthrough.	SEP SRAM macros	N/A
sep_sram_rsp_i	sep_pkg::sep_sram_rsp_t	Input	-	SEP scratch SRAM response passthrough.	SEP SRAM macros	N/A
sep_boot_rom_req_o	sep_pkg::sep_sram_req_t	Output	-	SEP boot ROM request passthrough.	SEP ROM macro	N/A
sep_boot_rom_rsp_i	sep_pkg::sep_sram_rsp_t	Input	-	SEP boot ROM response passthrough.	SEP ROM macro	N/A
sep_cpu_tcm_req_o	sep_pkg::sep_cpu_tcm_req_t	Output	-	SEP CPU TCM request passthrough (ICCM + DCCM).	SEP SRAM macros	N/A
sep_cpu_tcm_rsp_i	sep_pkg::sep_cpu_tcm_rsp_t	Input	-	SEP CPU TCM response passthrough.	SEP SRAM macros	N/A
sep_efuse_bank_ctrl_req_o	sep_efuse_pkg::efuse_axil_req_t	Output	-	SEP eFuse bank control AXI-Lite request passthrough.	SEP eFuse shim	N/A
sep_efuse_bank_ctrl_rsp_i	sep_efuse_pkg::efuse_axil_rsp_t	Input	-	SEP eFuse bank control AXI-Lite response passthrough.	SEP eFuse shim	N/A
sep_efuse_shim_command_req_o	sep_efuse_pkg::fuse_command_req_t	Output	-	SEP eFuse command interface request passthrough.	SEP eFuse shim	N/A
sep_efuse_shim_command_rsp_i	sep_efuse_pkg::fuse_command_rsp_t	Input	-	SEP eFuse command interface response passthrough.	SEP eFuse shim	N/A
sep_crypto_pkaimem_sram_req_o	sep_crypto_pkg::sep_crypto_pkaimem_sram_req_t	Output	-	SEP OTBN PKA instruction memory SRAM request passthrough.	SRAM macro	N/A
sep_crypto_pkaimem_sram_rsp_i	sep_crypto_pkg::sep_crypto_pkaimem_sram_rsp_t	Input	-	SEP OTBN PKA instruction memory SRAM response passthrough.	SRAM macro	N/A
sep_crypto_pkadmem_sram_req_o	sep_crypto_pkg::sep_crypto_pkadmem_sram_req_t	Output	-	SEP OTBN PKA data memory SRAM request passthrough.	SRAM macro	N/A
sep_crypto_pkadmem_sram_rsp_i	sep_crypto_pkg::sep_crypto_pkadmem_sram_rsp_t	Input	-	SEP OTBN PKA data memory SRAM response passthrough.	SRAM macro	N/A
sep_km_rom_mem_req_o	km_intf_pkg::km_rom_mem_req_t	Output	-	SEP Key Manager ROM memory request passthrough.	ROM macro	N/A
sep_km_rom_mem_rsp_i	km_intf_pkg::km_rom_mem_rsp_t	Input	-	SEP Key Manager ROM memory response passthrough.	ROM macro	N/A
sep_km_sram_mem_req_o	km_intf_pkg::km_sram_mem_req_t	Output	-	SEP Key Manager SRAM memory request passthrough.	SRAM macro	N/A
sep_km_sram_mem_rsp_i	km_intf_pkg::km_sram_mem_rsp_t	Input	-	SEP Key Manager SRAM memory response passthrough.	SRAM macro	N/A
ot_spi_irq_o	logic	Output	1	OT SPI interrupt output (to be muxed with external SPI controller).	SPI mux	N/A
spi_irq_i	logic	Input	1	Muxed SPI IRQ from sep_ip_integration .	SPI mux	Tie to '0 if unused

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
sep_external_req_o	sep_pkg::sep_32_64_6_12_axi_req_t	Output	-	SEP AXI extension port request passthrough.	Adopter peripherals	N/A
sep_external_resp_i	sep_pkg::sep_32_64_6_12_axi_resp_t	Input	-	SEP AXI extension port response passthrough.	Adopter peripherals	Tie to DECERR if unused
sep_reset_n_o	logic	Output	1	Plain SEP reset (post fuse-sense / JTAG override).	sep_ip_integration	N/A
sep_cpu_reset_n_o	logic	Output	1	SEP combined reset (sep_reset_n & WDT)	sep_ip_integration	N/A
sep_cpu_trace_o	sep_pkg::sep_cpu_trace_t	Output	-	SEP CPU trace output.	Trace infrastructure	N/A
lcc_demote_state_1_o	logic	Output	[1:0]	SEP lifecycle demote state 1.	External lifecycle logic	N/A
lcc_demote_state_2_o	logic	Output	[1:0]	SEP lifecycle demote state 2.	External lifecycle logic	N/A
sep_fuse_sense_done_o	logic	Output	1	SEP fuse sense done output.	System integration	N/A
clk_sep_wdt_i	logic	Input	1	SEP watchdog timer clock passthrough.	Low-frequency clock source	N/A
sep_straps_i	sep_pkg::sep_straps_t	Input	-	SEP configuration straps passthrough.	GPIO padring / tie-off	N/A

Chapter 2. System Management Network (SMN) Integration

WARNING

This section is currently under development. Content will be added to describe SMN integration procedures, network topology, and management interfaces.